CLINICAL QUANTUM ROADMAP • NHS LONG TERM CONDITIONS • SAMSKRIFT RESEARCH MONOGRAPH

Quantum-LTC: A Four-Era Clinical Quantum Optimisation and Simulation Roadmap Towards Fault-Tolerant Quantum Advantage for NHS Long Term Conditions

Arun Nadarasa[1,2,3]

[1]Innovation Fellowship Programme, Mid and South Essex NHS Foundation Trust, Basildon, Southend & Broomfield, UK
[2]International Social Prescribing Pharmacy Association (ISPPA), London, UK
[3]Primary Care & Integrated Neighbourhood Health Systems Research Group, London, UK

Portal & Live Space: arunnadarasa.github.io/quantum-ltc • hf.co/spaces/socialprescribing/quantum-ltc | **HF Model:** quantum-ltc-qltcmlm

Abstract. Managing multimorbid Long Term Conditions (LTCs) across primary care networks requires reconciling constrained clinic capacity, pre-winter high-risk stratification, polypharmacy safety, health-equity weighting, and decarbonisation targets [1, 2, 3]. Grounded in the **NHS North Central London Integrated Care Board (NCL ICB) Locally Commissioned Service (LCS) for Proactive, Personalised Care — Long Term Conditions (Version 6, 2026–2027)** [1], which unifies nine cardiometabolic, renal, hepatic, and respiratory conditions within a four-stage House-of-Care pathway [4, 5], we present **Quantum-LTC**: an end-to-end, cryptographically receipted clinical quantum optimisation and biophysical simulation roadmap spanning $N_{\text{logical}} \in [0, 1{,}000+]$ logical qubits. Crucially, we decouple *sparse discrete combinatorial scheduling*—where exact classical McCormick/Rosenberg mixed-integer linear programming (MILP) via HiGHS [6, 7, 8] dominates NISQ heuristics—from *strongly coupled open-system Lindbladian dynamics* and *metalloenzyme active-space quantum chemistry*, where full-density-matrix $\mathcal{O}(4^{n_{\text{spin}}})$ Krylov RAM walls, volume-law Matrix Product Density Operator (MPDO) bond growth [9], and $\mathcal{O}(1/\epsilon^2)$ Monte Carlo Wave Function (MCWF) trajectory variance [10] establish the earliest fault-tolerant crossover. Across three live **Guppy 1.0 → HUGR** jobs on **Quantinuum Nexus** `Helios-1E-lite` (6,144 emulator shots; 0 physical QPU shots), coherent Stage 1–4 multidisciplinary team (MDT) synthesis achieves $P(00) = 0.9961 \pm 0.0014$ versus 0.5020 ± 0.0110 under equal-unitary-depth siloed single-disease collapse, while an 8-rung mid-circuit measurement and reset (MCMR) recall ladder achieves $R^2 = 0.9979$ state preparation and measurement (SPAM) calibration linearity. On an 8-pathway 3-body cardiometabolic–renal–respiratory and Net-Zero dry-powder MART inhaler HUBO [11, 12], **Haiqu SDK** compresses $p = 10$ Linear-Ramp QAOA [13] depth from 223 → 26 (−88.34%), yielding 41.21% ± 1.54% raw ground-state probability (105.5× uniform random) and exact −23.500 ground energy. On a 128-variable NCL Call-and-Recall QUBO (**JijModeling/OpenJij**), raw simulated annealing (160/160 hard-constraint feasible, −113.85) plus 4-bit Kempe polish lifts pre-January High/Complex MDT recall from 14/18 (77.8%) to 18/18 (100.0%) within 0.80% (−134.57) of the 4.33 ms HiGHS MILP floor (−135.66). Biophysical, ab-initio electronic-structure, cloud QML, and governance modules integrate decoupled **HQS** `struqture-py/qoqo` 4-spin Lindbladians, **QProtSim** on `PDB 1N9V` [14], a paired **PySCF 2.15** [15] active-space benchmark contrasting closed-shell $3d^{10}$ `PDB 1O86` (`ACE`–lisinopril Zn^{2+}, $|E_{\text{RCCSD(T)}} - E_{\text{FCI}}| \leq 0.015$ mHa) [16] against open-shell hepatic `CYP3A4` Compound I (`PDB 1TQN`, where classical `RCCSD(T)` misses the Quartet–Doublet spin gap by 9.9–167.0 mHa, exact `FCI` exceeds 1 TiB at $M \geq 22$ orbitals, and Cholesky double-factorized qubitized QPE [17, 18] maps 26–58 orbitals onto 75–141 logical qubits) [19], a polarized `def2-SVP` C–H abstraction (`RC→TS_HAT`) benchmark on midazolam-bound `PDB 5TE8` [20] and ritonavir-bound `PDB 3NXU` [21] coupling spin-pure `CASSCF + SC-NEVPT2` (+10.17 vs. +12.16 kcal/mol [22, 23]), paired-barrier `THC-QPE` (2.56×–3.76× fewer Toffolis), and live **ChEMBL_37** (`CHEMBL340`) [24] Rowland–Matin PBPK `AUCR` closure, live **AlphaGenome 2** (`v2.dna_client`) primary-cell variant scoring [25, 26], 5-seed **Modal Cloud** `QLTCMLM` controls, **Rhyme** (`rhyme-beta`) [27], **Text2WetLab** OT-2 checks, **NIST FIPS 204** `ML-DSA-65` FHIR R4 provenance [28], **Kashefi Primitives Atlas v0.4** Lie-algebra $\mathfrak{g}$-sim audits [29, 30], and live **TypeSafe AI System One (`jev-latest` / Jev 1.14)** safety auditing under statutory **NHS DCB0129** and **DCB0160** [31, 32, 33].

Epistemic Demarcation & Statutory NHS DCB0129 / DCB0160 Clinical Safety Disclosure. All experimental results and ablations are deterministically reproducible from **eight SHA-256-sealed JSON receipts** in `results/` (`0e3216dd...`, `7212c3d9...`, `6ba83347...`, `cb1fec7c...`, `bd60742f...`, `a6b3854e...`, `1a9541a1...`, `164d38af...`) plus the 64-node knowledge graph (`1ee2e7f5...`). Every cohort ($N = 2{,}400$), PCN schedule ($P = 32$), HUBO tensor, and phenomenological Lindbladian rate is synthetic (`phi_present = False`; **0 physical QPU hardware shots**; **0 wet-lab human specimens**); in particular, the 75% cubic ($k = 3$) cohort interaction weighting is a stylized non-linear stress-test rather than an empirical fit to UK primary-care EHR distributions (e.g., CPRD Aurum) or validated NHS scores such as the Electronic Frailty Index (**eFI/eFI2**) [34] and **Cambridge Multimorbidity Score (CMS)** [35]. Classical HiGHS MILP, polynomial/kernel baselines, and single-reference `RCCSD(T)` (on closed-shell $3d^{10}$ `ACE` Zn^{2+}) dominate Era I–II and weakly correlated chemistry; fault-tolerant crossover is reserved strictly for Era III ($N_{\text{logical}} \in [38, 82]$) and Era IV ($N_{\text{logical}} \geq 100$) where open-shell transition-metal enzymes (`CYP3A4` Compound I, `PDB 1TQN/5TE8/3NXU`), bath-embedded volume-law Lindbladians, and dense frustrated $k \geq 4$ hypergraphs break classical `RCCSD(T)`, exact `FCI`/Krylov RAM, and bounded-χ tensor networks. Experimental structures from RCSB PDB (`CCO 1.0`), human microsomal bioactivities from EMBL-EBI `ChEMBL_37` (`CC BY-SA 3.0`), and AlphaGenome 2 functional annotations (governed by the AlphaGenome Non-Commercial Terms) are used strictly for non-commercial research benchmarking and never clinical diagnosis. All six clinical hazards (`H-LTC-01`–`H-LTC-06`) remain strictly **OPEN** across both manufacturer-side **NHS DCB0129** [32] and deploying-organisation **NHS DCB0160** [33] governance under mandatory GP, Clinical Pharmacist, and Tier 2/3 Specialist Nurse sign-off.

1. Clinical Specification: The NHS NCL ICB Multimorbid LTC Service Model

Multimorbidity—the co-occurrence of two or more chronic conditions within an individual—is projected by the Health Foundation REAL Centre to expand substantially across England through 2040, with severe illness concentrated disproportionately in the most socioeconomically deprived quintile [2]. Traditional Quality and Outcomes Framework (QOF) single-disease templates frequently subject multimorbid patients to fragmented, duplicative clinic appointments, conflicting polypharmacy adjustments, and uncoordinated lifestyle advice [3, 4]. To overcome single-condition fragmentation across Barnet, Camden, Enfield, Haringey, and Islington, the **NHS North Central London Integrated Care Board (NCL ICB)** commissioned Version 6 of the *Locally Commissioned Service (LCS) for Proactive, Personalised Care — Long Term Conditions* (1 April 2026 – 31 March 2027) [1].

The NCL LTC LCS unifies **nine conditions** accounting for approximately 75% of modifiable cardiometabolic, renal, hepatic, and respiratory morbidity: **(1)** Cardiovascular Disease (coronary heart disease, stroke/TIA, peripheral arterial disease, heart failure), **(2)** Atrial Fibrillation, **(3)** Hypertension (excluding low-risk well-controlled cases, anchored by **NCL PCN Outcome #23**: achieving blood pressure $< 140/90$ mmHg [home $< 135/85$] for age < 80 or $< 150/90$ mmHg [home $< 145/85$] for age ≥ 80 across all five boroughs), **(4)** Type 2 Diabetes, **(5)** Type 1 Diabetes (mandated `High Risk + Complex`), **(6)** Chronic Kidney Disease (CKD), **(7)** Non-Alcoholic Fatty Liver Disease / Metabolic Dysfunction-Associated Steatotic Liver Disease (NAFLD/MASLD), **(8)** COPD, and **(9)** Adult and Children/Young People (CYP, aged 0–17) Asthma, which mandates a **48-hour post-attack review** (within two working days of an emergency or out-of-hours presentation) and a **6-week full review** by Tier 2/3 trained clinicians alongside short-acting β_2-agonist (SABA)-free pathways [1, 12].

Operationally, the specification embeds the four-stage **Year of Care / House of Care** cycle [4, 5]:

1. **Proactive Case Finding & Call-and-Recall Tension (Page 5 of LCS v6):** Practices run 3-yearly UCLPartners (UCLP) searches to close undiagnosed prevalence gaps and face an explicit scheduling dilemma between *Month-of-Birth recall* (which smooths staffing across 12 months but leaves high-risk patients exposed to winter admissions) and *Risk-Stratified recall* (`Low`, `Moderate`, `High`, `High Risk + Complex`), which prioritizes pre-winter proactive care (capable of reducing emergency admissions by up to 38% and GP contacts by 18% [5]) at the risk of seasonal MDT bottlenecks. In NHS GP EHRs (EMIS Web / SystmOne / Medicus), this 4-tier operational stratification complements validated primary-care frailty and multimorbidity instruments—specifically the cumulative-deficit **Electronic Frailty Index (eFI/eFI2)** [34] and the consultation/admission/mortality-weighted **Cambridge Multimorbidity Score (CMS)** [35]—by coupling longitudinal diagnostic deficits with acute physiological biomarkers (Outcome #23 BP, uACR/eGFR, HbA1c, post-attack CYP asthma) and linguistic/deprivation equity weights.

2. **Stage 1 (Check and Test) & Stage 2 (Re-Stratification & MDT Desktop Review):** A single 30-minute baseline appointment gathers phlebotomy (HbA1c, eGFR, lipids, LFTs/FIB-4, NT-proBNP), urine albumin-creatinine ratio (uACR), blood pressure, foot checks, and SNOMED CT `315594003` (`Interpreter needed`) access requirements, followed by Clinical Pharmacist / MDT Desktop Review for `High` and `High + Complex` patients.

3. **Stage 3 (PCSP Discussion) & Stage 4 (Continuity Follow-Up):** Two to four weeks after test results are shared, a 30-minute Coaching-for-Health appointment co-produces a Personalised Care and Support Plan (PCSP) integrating pharmacotherapy, Voluntary, Community and Social Enterprise (VCSE) social prescribing [36], and **NCL Green Plan Net Zero** switches (preventing carbon-intensive haemodialysis via early SGLT2i/ACEi protection [11] and replacing high-global-warming-potential propellant SABA inhalers with dry-powder Maintenance and Reliever Therapy [MART] [12]), followed by a clinician-continuity Stage 4 review evaluated via `collaboRATE` (≥ 25 Top-Score responses) and monthly prospective **PCN Weighted Payments** targeting the 20% most deprived areas (IMD Quintile 1) and minority ethnic communities [1].

Table 1: Four-Era Clinical Quantum Optimisation and Simulation Roadmap for NHS Long Term Conditions (N_{logical} Scaling). Physical qubit footprints follow $N_{\text{phys}} = (\lceil 1.5 N_{\text{logical}} \rceil + 12) \times 2d^2$ incorporating rotated surface-code tiles, lattice-surgery routing (1.5×), and a 12-tile magic-state factory ($p_{\text{phys}} = 10^{-3}$) [40, 41, 38]. Memory sizes use strict binary units (1 GiB = 2^{30} B, 1 TiB = 2^{40} B).

Roadmap Era	Logical Qubits	Physical Qubits	QEC & Hardware Regime	Target NHS NCL LTC Clinical & Biophysical Workloads	Classical vs. Quantum Verdict
Era I *Classical-Exact & NISQ Readiness*	0 logical (1–20 phys / cloud emul.)	1–98 (Helios / NISQ)	Uncorrected physical gates + 8-rung affine SPAM calibration	128-var PCN Call/Recall scheduling QUBO; 8-var 3-body multimorbid + Net-Zero MART HUBO; 4q Frobenius `QLTCMLM`	**Classical Dominates** (HiGHS MILP < 25 ms; 0 runtime speedup claimed)
Era II *Error-Detected Logical*	3–24 error-detected	10–192 (Helios / Sol QCCD)	Distance-2 $[\![8,3,2]\!]$ & $[\![4,2,2]\!]$ color codes + post-selection [38]	Shallow MCMR longitudinal recall combs (CYP Asthma 48h/6wk & BP #23); compressed $p \leq 3$–10 LR-QAOA circuits	**Classical MPS/MILP Faster** (Post-selection yield $p_{\text{acc}} \to 15.8\%$ at 24L, $d_{\text{circ}} = 24$)
Era III *Early Active FTQC Crossover*	24–75 active QEC	5,300–90,250 ($d = 9$–19 + T-factory)	Active multi-round decoding + magic-state cultivation / distillation	Open-shell `CYP3A4` Cpd I `CAS(27e,26o)` (75L, `1TQN`) vs. single-ref `ACE` Zn^{2+} (`1O86`); 16–24 spin organ Lindbladians (38–54L)	**First Simulation Crossover** (`RCCSD(T)` fails `1TQN`; FCI > 1 TiB at $M \geq 22$)
Era IV *Asymptotic ICS-Scale FTQC*	100–1,000+ active QEC	90,000–250,000+ ($d = 15$–21 modular)	Photonic-linked QCCD + parallel magic-state factories	Drug-bound `CYP3A4` (`5TE8`/`3NXU`) 58o–76o DF/THC-QPE (140–186L); QSVT organ twins (128L) [42]; $\mathcal{O}(1/\epsilon)$ QAE CVaR [43]; Quantum B&B [44]	**Projected FTQC Advantage** (Advisory only under NHS DCB0129 & DCB0160 sign-off)

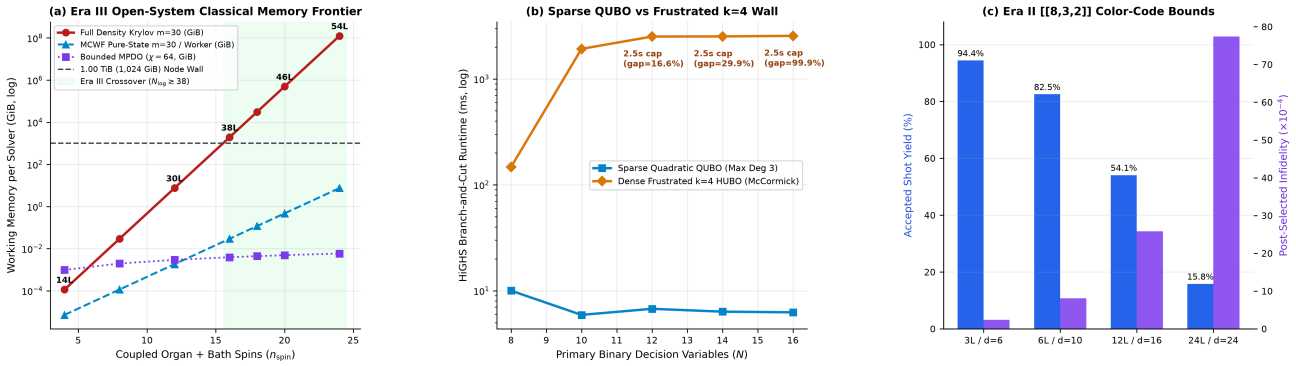


Figure 1: Quantitative Scaling Laws Across the Four-Era Logical Qubit Roadmap (`modal_roadmap_rhyme_qml_receipt.json` & `deep_review_remediation_receipt.json`). **(a)** Open-system n_{spin} Lindbladian memory in strict binary gibibytes (GiB = 2^{30} B): exact $m = 30$ full-density-matrix Krylov integration ($\dim \mathcal{L} = 4^{n_{\text{spin}}}$) crosses 1 TiB at $n_{\text{spin}} = 16$ (1,920 GiB = 1.875 TiB, 38 logical qubits) and reaches 480.0 TiB at $n_{\text{spin}} = 20$ (46 logical / 19,602 physical qubits at $d = 11$), contrasted against single-trajectory MCWF pure-state Krylov memory (30 MiB at $n_{\text{spin}} = 16$, 480 MiB at $n_{\text{spin}} = 20$, trading RAM for $\mathcal{O}(1/\epsilon^2)$ trajectory sampling [10]) and bounded ($\chi = 64$) vs. volume-law MPDOs [9]. **(b)** Empirical HiGHS branch-and-cut runtime comparing sparse quadratic QUBOs (max vertex degree 3; 6.26 ms at $N = 16$) against dense frustrated quartic ($k = 4$) HUBOs, which hit the 2.5 s cap from $N = 12$ upward with widening percentage MIP gaps (16.6% → 29.9% → 99.9%). **(c)** Analytical Era II $[\![8,3,2]\!]$ color-code post-selection shot yield $p_{\text{acc}} = (1 - p_{2q})^{N_{\text{loc}}}$ and post-selected logical infidelity $p_{\text{log,eff}} = \min(1, 3.5 N_{\text{loc}} p_{2q}^2)$ at $p_{2q} = 1.2 \times 10^{-3}$.

2. The Four-Era Logical Qubit Roadmap: Decoupling Optimization from Simulation

A pervasive pitfall in translational quantum healthcare proposals is conflating the exponential operator-space growth of strongly entangled open quantum systems with sparse discrete combinatorial scheduling, which modern classical branch-and-cut solvers handle effortlessly via polynomial linearization [6, 7, 8]. To establish an intellectually rigorous trajectory as hardware progresses from noisy intermediate-scale quantum (NISQ) processors to modular trapped-ion fault tolerance [37, 38, 39], we partition clinical workloads across **four logical-qubit eras** (Table 1 and Figure 1).

2.1 Mathematical Scaling Separation: McCormick MILP, Trajectory/MPDO Baselines & FTQC

Consider first a general k-local clinical decision hypergraph over N binary interventions $x \in \{0,1\}^N$:

$$E_{\mathrm{HUBO}}(x) = \sum_{i=1}^{N} h_i x_i + \sum_{(i,j)\in E_2} J_{ij} x_i x_j + \sum_{(i,j,\ell)\in E_3} K_{ij\ell} x_i x_j x_\ell + \sum_{(i,j,\ell,m)\in E_4} M_{ij\ell m} x_i x_j x_\ell x_m. \tag{1}$$

Introducing continuous auxiliary variables $z_S = \prod_{i\in S} x_i \in [0,1]$ for each hyperedge S of order $|S| = k$ with exact McCormick linear inequalities [7]:

$$z_S \le x_i \quad (\forall i \in S), \qquad z_S \ge \sum_{i\in S} x_i - (k-1), \qquad z_S \ge 0, \tag{2}$$

reduces Eq. (1) to an exact MILP with N binary variables, $|E_{\ge 2}|$ continuous variables, and $\sum_S(|S|+1)$ structural linear constraint rows (since $z_S \ge 0$ is enforced directly via the solver box bound $z_S \in [0,1]$, yielding $5\times 1{,}820 = 9{,}100$ rows at $N = 16, k = 4$). When the interaction hypergraph is sparse, dual-simplex + Gomory/MIR cutting planes in HiGHS [8] close the root LP gap in milliseconds. Under minimization ($\min_x E_{\mathrm{HUBO}}(x)$) on dense frustrated instances containing competing positive and negative couplings, however, the fractional midpoint $x_i = 1/2$ satisfies $\sum_{i\in S} x_i - (k-1) = 1 - k/2 \le 0$ for all $k \ge 2$ (strictly negative for $k \ge 3$): the LP relaxation simultaneously sets $z_S = 0$ freely on *positive-cost* (repulsive/penalty) hyperedges ($M_S > 0$, zeroing out frustration penalties even though $\prod_{i\in S} x_i = 2^{-k} > 0$) while setting $z_S = 1/2 \gg 2^{-k}$ on *negative-cost* (attractive/reward) hyperedges ($M_S < 0$, over-collecting rewards by $8\times$ at $k = 4$). Empirically (Figure 1b), while sparse quadratic QUBOs (maximum graph vertex degree 3) solve in 6.26 ms at $N = 16$, dense frustrated $\binom{N}{4}$ quartic HUBOs jump from 147.3 ms ($N = 8$, 70 terms) and 1,920.3 ms ($N = 10$, 210 terms) to hit the 2.5 s branch-and-cut cap at $N = 12, 14, 16$ (1,820 terms; residual MIP integrality gap 16.6% → 29.9% → 99.9%), motivating Era IV quantum branch-and-bound ($\mathcal{O}(\sqrt{T_{\mathrm{tree}}})$) [44].

Conversely, consider open-system dynamics generated by a Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation over n_{spin} two-level degrees of freedom,

$$\frac{d\rho}{dt} = \mathcal{L}(\rho) = -i[H_{\mathrm{organ}}, \rho] + \sum_{\mu} \gamma_\mu \left(L_\mu \rho L_\mu^\dagger - \frac{1}{2}\{L_\mu^\dagger L_\mu, \rho\} \right). \tag{3}$$

Mathematically, Eq. (3) with constant rates $\gamma_\mu \ge 0$ generates a strictly Markovian (CP-divisible) semigroup on the full n_{spin} register; non-Markovian memory-kernel dynamics on n_{organ} physiological compartments arise when $n_{\mathrm{spin}} = n_{\mathrm{organ}} + n_{\mathrm{bath}}$ embeds explicit environmental/neurohumoral bath pseudomodes that are subsequently traced out ($\rho_{\mathrm{organ}}(t) = \mathrm{Tr}_{\mathrm{bath}}\rho(t)$). Benchmarking quantum simulation fairly requires comparing against **three distinct classical open-system paradigms** (Figure 1a):

1. **Exact Full-Density-Matrix Krylov Integration ($\mathcal{O}(4^{n_{\mathrm{spin}}})$):** Storing $m = 30$ double-precision complex vectors (16 B/entry) in $\mathbb{C}^{4^{n_{\mathrm{spin}}}}$ requires $30 \times 16 \times 4^{n_{\mathrm{spin}}}$ bytes: in strict binary units, 7.50 GiB at $n_{\mathrm{spin}} = 12$, crossing 1 TiB at $n_{\mathrm{spin}} = 16$ (**1,920** GiB = **1.875** TiB), **30.00** TiB at $n_{\mathrm{spin}} = 18$, and **480.00** TiB at $n_{\mathrm{spin}} = 20$ (527.77 TB decimal SI).

2. **Monte Carlo Wave Function (MCWF) Trajectory Unraveling [10]:** Propagating pure statevectors $|\psi(t)\rangle \in \mathbb{C}^{2^{n_{\mathrm{spin}}}}$ interleaved with stochastic jumps reduces working memory to $\mathcal{O}(2^{n_{\mathrm{spin}}})$—only 1.00 MiB (30.0 MiB for $m = 30$ Krylov) at $n_{\mathrm{spin}} = 16$ and 16.00 MiB (480.0 MiB Krylov) at $n_{\mathrm{spin}} = 20$—eliminating the single-node RAM wall at the expense of $\mathcal{O}(1/\epsilon^2)$ Monte Carlo trajectory variance ($\approx 10^6$ independent trajectories per parameter point to resolve $\epsilon = 10^{-3}$ steady-state shifts or rare adverse-event tails, compared with coherent $\mathcal{O}(1/\epsilon)$ QSVT/QAE scaling).

3. **Matrix Product Density Operators (MPDOs) [9]:** Locally purified tensor networks compress 1D/area-law dissipative chains into $\mathcal{O}(n_{\mathrm{spin}}\chi^2)$ memory (≤ 5.0 MiB at $\chi = 64, n_{\mathrm{spin}} = 20$), remaining faster than Era II–III hardware whenever operator-space entanglement stays bounded, but encountering exponential bond growth ($\chi \sim 2^{n_{\mathrm{spin}}}$) when non-planar multi-organ crosstalk and strongly coupled bath pseudomodes generate volume-law operator entanglement.

Under Quantum Singular Value Transformation (QSVT) block-encoding [42], an $n_{\mathrm{spin}} = 20$ bath-embedded Lindbladian requires $N_{\mathrm{logical}} = 2n_{\mathrm{spin}} + 6 = 46$ active logical qubits (40 Stinespring-dilated system/purification qubits, 4 linear-combination-of-unitaries [LCU] prepare/select ancillas indexing the jump operators L_μ, and 2 QSVT signal-processing ancillas). At an assumed T-gate budget of 7.2×10^4 ($p_{\mathrm{phys}} = 10^{-3}$), rotated surface-code distance $d = 11$ ($2d^2 = 242$ physical qubits/tile) under

$$N_{\mathrm{phys}}(N_{\mathrm{logical}}, d) = (\lceil 1.5\, N_{\mathrm{logical}} \rceil + N_{\mathrm{factory}}) \times 2d^2, \qquad N_{\mathrm{factory}} = 12, \tag{4}$$

yields $(69 + 12) \times 242 =$ **19,602** physical qubits, pinpointing **Era III (**38–68 **active logical qubits)** as the earliest crossover against exact Liouvillian and volume-law MPDO/chemistry regimes.

In **Era II (3–24 error-detected logical qubits)**, $B = N_{\text{logical}}/3$ blocks of the $[\![8,3,2]\!]$ color code ($n_{\text{phys}} = 8B$) [38] accrue $N_{\text{loc}} = 8B\,d_{\text{circ}} = n_{\text{phys}} d_{\text{circ}}$ effective qubit-layer fault locations across circuit depth d_{circ} (treating $p_{2q} = 1.2 \times 10^{-3}$ as an effective per-qubit-per-layer fault rate folding two-qubit gate, syndrome-extraction, and idling errors). Trivial-syndrome post-selection yields survival $p_{\text{acc}} = (1 - p_{2q})^{N_{\text{loc}}}$ and undetected weight-≥ 2 logical infidelity $p_{\text{log,eff}} = \min(1,\, 3.5 N_{\text{loc}} p_{2q}^2)$: at 3L ($B = 1, d_{\text{circ}} = 6, N_{\text{loc}} = 48$), $p_{\text{acc}} = 94.40\%$ and $p_{\text{log,eff}} = 2.42 \times 10^{-4}$, whereas at 24L ($B = 8, d_{\text{circ}} = 24, N_{\text{loc}} = 1{,}536$), survival drops to 15.81% (6.32× shot overhead; Figure 1c), necessitating active Era III syndrome decoding beyond $N_{\text{logical}} \approx 24$.

3. Live Trapped-Ion Emulation & Combinatorial Optimisation (Era I–II)

3.1 Live Quantinuum Nexus `Helios-1E-lite` Guppy 1.0 → HUGR Execution

We compiled three clinical verification circuits in **Guppy 1.0** directly to Quantinuum's Hierarchical Unified Graph Representation (**HUGR**) and executed them live on **Quantinuum Nexus `Helios-1E-lite`** (`QSystemErrorModel("alpha")`, project `Quantum-LTC-NCL`, ID `69eb4683-8de0-4dc9-bd22-ce6d98cafc7f`; $2{,}048 \times 3 = 6{,}144$ live cloud emulator shots; 0 physical QPU shots; Figure 2a–b):

1. **Equal-Unitary-Depth House-of-Care MDT Synthesis vs. Siloed Collapse** (Job UUID `5da15d5a-c48e-4a11-9bb3-6c0d8854131c`, HUGR `7,456 B`): Evaluates two 2-qubit arms simultaneously within a 4-qubit register. Acting *right-to-left* on $|00\rangle$ (so $(H \otimes I)$ and CX_{01} prepare Stage 1 Bell entanglement $|\Phi^+\rangle$ before local clinical-axis rotations and Stage 2 SWAP handoff), Arm A (q_0, q_1) applies $U_{\text{MDT}} = (H \otimes I)\,\mathrm{CX}_{01}\,(R_z(-\theta_z) \otimes R_y(-\theta_y))\,\mathrm{SWAP}_{01}\,(R_y(\theta_y) \otimes R_z(\theta_z))\,\mathrm{CX}_{01}\,(H \otimes I)$ at $(\theta_y, \theta_z) = (0.38\pi, 0.28\pi)$. Arm B (q_2, q_3) executes the *exact same 1Q/2Q unitary gates* interrupted immediately *after* $\mathrm{SWAP}(q_2, q_3)$ and *before* $(R_z(-\theta_z) \otimes R_y(-\theta_y))$ by a premature siloed projective measurement `project_z(q2)`: algebraically, $\mathrm{SWAP}(q_2, q_3)$ routes the Z-diagonal branch $R_z(\theta_z)|b\rangle$ onto q_2 and $R_y(\theta_y)|b\rangle$ onto q_3, so projecting q_2 destroys inter-stage Bell coherence while preserving pure $R_y(\theta_y)|b\rangle$ on q_3, which $R_y(-\theta_y)_{q_3}$ still inverts to $|b\rangle$, mapping CX_{23} and H_{q_2} identically to $\rho_{\text{silo}} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|10\rangle\langle 10|$ ($P_{\text{silo}}(00) = 0.5000$ across all (θ_y, θ_z), whereas projecting prior to SWAP would yield $\frac{1}{2}\cos^2\theta_y = 0.0677$). Defining $H_1 = -\sum_{x \in \{0,1\}^2} P(x) \log_2 P(x)$ as the 4-outcome marginal Shannon entropy over $\{00, 01, 10, 11\}$, live 2,048-shot counts are $(n_{00}, n_{01}, n_{10}, n_{11}) = (\mathbf{2040}, \mathbf{1}, \mathbf{3}, \mathbf{4})$ for Arm A—giving $P_{\text{MDT}}(00) = \mathbf{0.9961 \pm 0.0014}$ (1σ binomial SE) and $H_1 = 0.0424$ bits—versus $(\mathbf{1028}, \mathbf{1}, \mathbf{1014}, \mathbf{5})$ for Arm B, giving $P_{\text{silo}}(00) = \mathbf{0.5020 \pm 0.0110}$ (0.18σ from 0.5000) and $H_1 = 1.0278$ bits (> 1 bit solely from $6/2{,}048 = 0.29\%$ physical gate/SPAM leakage into $\{01, 11\}$), matching the 982.4 μs classical 4×4 density-matrix floor.
2. **8-Rung Mid-Circuit Recall & CYP Asthma 48h Calibration Ladder** (Job UUID `464be33a-6ca0-4d6d-af99-2d7eba06ce8a`, HUGR `7,566 B`): Executes eight sequential `project_z` + `reset` rungs on one trapped-ion qubit, preparing rung r via $R_y(\pi \cdot \mathrm{HT}_r)$ with half-turn angles $\mathrm{HT}_{1..5} = (0.6667, 0.5804, 0.5000, 0.4196, 0.3333)$ for SPAM standards $\langle Z \rangle \in \{-0.50, -0.25, 0.00, +0.25, +0.50\}$ and $\mathrm{HT}_{6..8} = (0.3620, 0.2339, 0.5000)$ for clinical recall profiles. Ordinary least-squares fit $\hat{C} = 0.9761\,C_{\text{raw}} - 0.0114$ achieves $\mathbf{R^2 = 0.9979}$, recovering $+0.7150 \pm 0.0156$ calibrated retention contrast for proactive pre-winter + CYP 48h/6-week cadence versus $+0.3727 \pm 0.0211$ for isolated Month-of-Birth review and $+0.0057 \pm 0.0226$ on the null twin.
3. **4-Qubit Frobenius `QLTCMLM` Inference Circuit** (Job UUID `0a1fb43e-4998-4700-b9ec-4f6b0b73e677`, HUGR `6,734 B`): Executes a 2-layer, 8-two-qubit-gate (4 cyclic CX + 2 diagonal CZ in Layer 1; 2 cross CX in Layer 2; 24 single-qubit $R_y/R_z/R_x$ rotations) evaluation of a representative high-risk cardiorenal–asthma profile, achieving Bhattacharyya fidelity **0.9990** (TVD = 0.0287) against the exact statevector.

3.2 Haiqu 8-Qubit Multimorbid Regimen & Net-Zero Inhaler 3-Body HUBO

To co-optimise cardiometabolic, renal, respiratory, and NCL Green Plan Net-Zero priorities [1, 11, 12], we formulated an 8-variable cubic HUBO (x_0: ACEi/ARB titration to Outcome #23 target; x_1: SGLT2i cardiorenal–MASLD protection; x_2: SABA-free dry-powder MART inhaler switch; x_3: Pharmacist SMR deprescribing nephrotoxic NSAIDs; x_4: Stage 3 PCSP coaching + VCSE referral; x_5: Home BP $< 135/85$ telemonitoring; x_6: retain high-GWP SABA + NSAID status quo; x_7: fragmented duplicate single-disease clinics) with linear biases $h = (-2.5, -2.3, -2.1, -2.4, -1.8, -1.6, +2.9, +3.1)$, nonzero pairwise couplings $J_{06} = +6.2, J_{26} = +5.4, J_{47} = +4.8, J_{05} = -1.4, J_{13} = -1.3, J_{34} = -1.1$ (all unlisted $J_{ij} = 0$), and four 3-body cubic terms:

$$H_{\text{cubic}}(x) = -3.9\,x_0x_1x_3 - 3.1\,x_2x_4x_5 + 7.2\,x_0x_1x_6 + 3.4\,x_1x_4x_7, \tag{5}$$

where $+7.2\,x_0x_1x_6$ penalizes the acute kidney injury (AKI) "triple-whammy" interaction (ACEi/ARB + SGLT2i hemodynamic shift + NSAID) and $-3.9\,x_0x_1x_3$ rewards Pharmacist NSAID-free cardiorenal protection. Compiling a $p = 10$-layer Linear-Ramp QAOA circuit [13] with ramp slope $\Delta\gamma = \Delta\beta = 0.5$ (`delta=0.5`, decomposing cubic terms via ancilla-free 6-CNOT ZZZ phase gadgets over $\{\mathrm{CX}, R_z, R_x\}$) yields baseline depth 223 (164 routed 2Q gates). Applying **Haiqu SDK** `state_compression`—an approximate fixed-initial-state ($|0\rangle^{\otimes 8}$) tensor-network synthesizer (`approximation_level=2`, achieving noiseless state overlap $|\langle\psi_{\text{orig}}|\psi_{\text{comp}}\rangle|^2 = 0.9581$)—compresses circuit depth from **223** → **26** (**88.34**% reduction; 2Q gates 164 → 27). On noiseless `aer_simulator` (1,024 shots), **raw pre-postprocessing** sampling achieves mean energy -14.530 ± 0.292, $\mathrm{CVaR}_{0.10} = -23.500$, and samples the

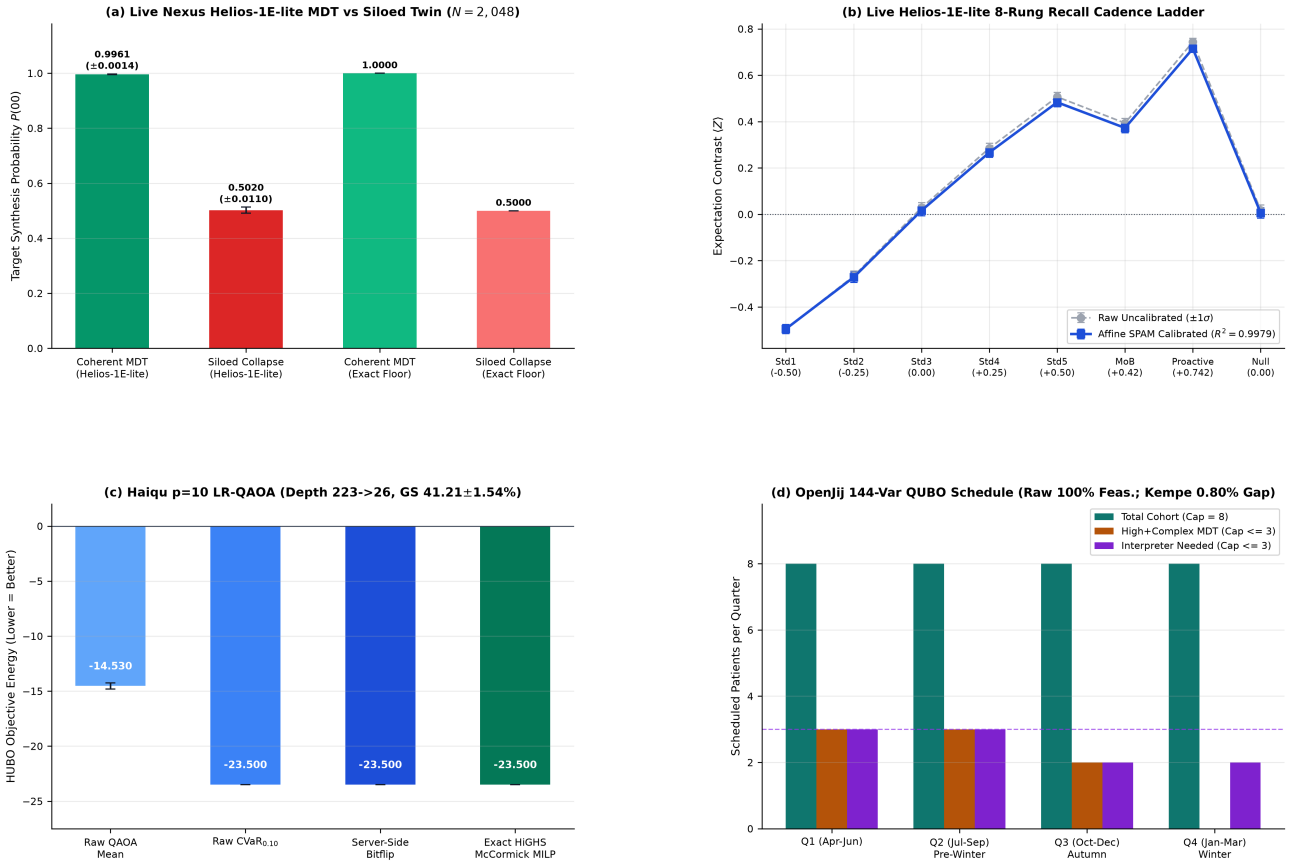


Figure 2: Live Quantinuum Nexus `Helios-1E-lite`, Haiqu 3-Body HUBO, and OpenJij NCL Call-and-Recall Benchmarks (`nexus_helios_ltc_receipt.json`, `haiqu_jij_ltc_receipt.json`, & `deep_review_remediation_receipt.json`; error bars denote $\pm 1\sigma$ binomial/shot SE). **(a)** Equal-unitary-depth Stage 1–4 MDT synthesis ($P(00) = 0.9961 \pm 0.0014$) versus post-SWAP siloed collapse (0.5020 ± 0.0110). **(b)** 8-rung mid-circuit recall ladder ($R^2 = 0.9979$) across five SPAM standards $\{-0.50, -0.25, 0.00, +0.25, +0.50\}$ and three clinical recall profiles. **(c)** Haiqu SDK 8-qubit $p = 10$ LR-QAOA 3-body HUBO showing raw QAOA mean (-14.530 ± 0.292, raw GS hit rate $41.21\% \pm 1.54\%$), $\text{CVaR}_{0.10}$ (-23.500), server-side bitflip postprocessing (-23.500), and exact HiGHS MILP floor (-23.500). **(d)** OpenJij + 4-bit Kempe-swap 128-variable NCL PCN Call-and-Recall schedule satisfying exact 8/8/8/8 clinic smoothing, ≤ 3 MDT, and ≤ 3 `Interpreter needed` caps while scheduling 18/18 (100%) High/Complex patients before January.

unique ground bundle `11111100` ($E_0 = -23.500$, invariant across 100% of a 25-point $(\alpha_{\text{syn}}, \beta_{\text{haz}}) \in [0.6, 1.4]^2$ grid) on $\mathbf{41.21\% \pm 1.54\%}$ (422/1,024) of raw shots (105.5× uniform random; an independent uncompressed $d = 223$ vs. compressed $d = 26$ paired run yields $36.82\% \pm 1.51\%$ vs. $38.67\% \pm 1.52\%$, consistent within 1.2σ). Crucially, exhaustive landscape analysis over all $2^8 = 256$ states confirms that `11111100` is the unique strict 1-Hamming local minimum: 5-pass greedy 1-bitflip seeded from a *single* uniform random bitstring already reaches -23.500 with 85.55% probability (219/256 states; 100% best-of-1,024 under `haiqu.postprocess`), confirming that QAOA's contribution on this 8-variable benchmark lies in its 105.5× raw pre-postprocessing state amplification rather than postprocessed convergence, while classical 129.9 μs enumeration and 23.03 ms HiGHS MILP remain faster at Era I scale (Figure 2c). Haiqu **Entangled Manifold Embedding** encoded an 8-feature profile into 4 qubits in live cloud execution, while a 4-qubit SU(2)-equivariant brickwork ansatz verified zero spin-symmetry violation ($0.0 < 10^{-15}$) and **Observable Backpropagation** tracked cardiorenal observables cleanly.

3.3 JijModeling & OpenJij 128-Variable NCL Call-and-Recall Scheduling QUBO

Directly formalizing Page 5 of the NCL LTC LCS v6 specification [1], we scheduled $P = 32$ representative PCN patients (8 `High_Complex_MDT`, 10 `High_Risk`, 8 `Moderate_Risk`, 6 `Low_Risk`; birth quarter $b_p = p \bmod 4$; 10 requiring SNOMED `Interpreter needed` extended slots $\mathcal{I}$; 14 carrying equity weight $w_{\text{eq}}(p) = 1.45$ vs. 1.00) across

$Q = 4$ quarters (128 primary binary variables $x_{p,q}$ plus 16 binary slack variables s):

$$\min_{x\in\{0,1\}^{128},\, s\in\{0,1\}^{16}} \sum_{p=0}^{31}\sum_{q=0}^{3} C_{p,q}x_{p,q} + \lambda_1 \sum_{p=0}^{31}\Big(\sum_{q=0}^{3} x_{p,q} - 1\Big)^2 + \lambda_2 \sum_{q=0}^{3}\Big(\sum_{p=0}^{31} x_{p,q} - 8\Big)^2 + \lambda_3 \sum_{q=0}^{3}\Big(\sum_{p\in\mathcal{M}} x_{p,q} + s^{\mathrm{mdt}}_{q,0} + 2s^{\mathrm{mdt}}_{q,1} - 3\Big)^2 + \lambda_4 \sum_{q=0}^{3}\Big(\sum_{p\in\mathcal{I}} x_{p,q} + s^{\mathrm{int}}_{q,0} + 2s^{\mathrm{int}}_{q,1} - 3\Big)^2, \tag{6}$$

with $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (30, 12, 10, 10)$. The linear priority cost matrix is explicitly $C_{p,q} = -w_{\mathrm{eq}}(p)\,[S(\mathrm{tier}_p, q) + B_{\mathrm{MoB}}(\mathrm{tier}_p)\,\mathbb{I}(q = b_p) + \xi_{p,q}]$ ($\xi_{p,q} \sim \mathcal{U}(-0.08, 0.08)$, `seed=202604`), with seasonal pre-winter utility $S(\mathrm{High_Complex}) = (3.1, 4.2, 3.8, 1.1)$, $S(\mathrm{High}) = (2.8, 3.9, 3.6, 1.4)$, $S(\mathrm{Mod}) = 2.0$, $S(\mathrm{Low}) = 1.6$ and Month-of-Birth alignment bonus $B_{\mathrm{MoB}} = (0.35, 0.45, 1.65, 2.10)$, giving $C_{p,q} \in [-6.2039, -1.0944]$ and $\max_{p,q,q'} |C_{p,q} - C_{p,q'}| = 4.6580 < \min_k \lambda_k = 10.0$ (strictly preventing any hard-constraint violation from trading favorably against linear priority gain). Separating raw annealing from classical local search: **raw pre-polish OpenJij `SASampler`** (160 reads, 2,500 sweeps, standalone runtime 417.81 ms) already achieves **100.0**% (**160/160** reads) hard-constraint feasibility across one-hot, 8/8/8/8 clinic, ≤ 3 MDT, and ≤ 3 interpreter caps with raw feasible linear objective -113.8488 (16.08% gap vs. MILP), because single-spin Metropolis flips between feasible schedules must cross $\Delta E \approx \lambda_1 + \lambda_2 = +42$ penalty barriers. Applying a lightweight 4-bit Kempe-chain swap polish (508.78 ms total pipeline runtime) tunnels across equal-capacity patient swaps to reach -134.5719—within **0.80**% of the exact HiGHS MILP floor (-135.6623 solved in 4.33–32.79 ms; Figure 2d)—lifting pre-January High/Complex recall from 14/18 (77.8%) to **18/18** (**100.0**%) while maintaining quarterly MDT counts (`cap` ≤ 3; quarterly counts: $3, 3, 2, 0$) and interpreter counts (`cap` ≤ 3; quarterly counts: $3, 3, 2, 2$).

4. Open-System Biophysical, Structural, and Genomic Grounding (Era II–III)

4.1 Phenomenological HQS 4-Spin Organ Lindbladian, Decoupled Ablation & `QProtSim`

We emphasize upfront that modeling macroscopic organ compartments (0: vascular RAAS tone [`ACE`], 1: glycaemic/MASLD hepatic axis [`GLP1R`], 2: renal tubuloglomerular filtration [`SLC5A2`/SGLT2], 3: airway/IL-6 inflammation [`ADRB2`/`IL6`]) as coupled quantum spins is a **stylized phenomenological benchmark for open-system solver scaling**, not a mechanistic biophysical claim of macroscopic quantum phase coherence across human organs (where classical nonlinear SDEs and continuous-time Markov chains remain standard). In **HQS Quantum Simulations' `struqture-py` and `qoqo`**, we specify:

$$H_{\mathrm{LTC}} = \sum_{i=0}^{3} \frac{\omega_i}{2} Z_i + \sum_{(i,j)\in\mathcal{E}} g_{ij}(X_iX_j + Y_iY_j) + \sum_{i=0}^{3} \Omega_i X_i, \tag{7}$$

with $\omega/2 = (0.42, 0.36, 0.40, 0.46)$, coupling graph $\mathcal{E} = \{(0,2) : 0.44,\ (1,2) : 0.34,\ (0,1) : 0.28,\ (0,3) : 0.26,\ (2,3) : 0.24\}$, baseline drive $\Omega_{\mathrm{unc}} = (0.05, 0.00, 0.05, 0.00)$, and proactive drive $\Omega_{\mathrm{pro}} = (0.58, 0.46, 0.60, 0.52)$, initialized at decompensated $|1111\rangle$ and evolved to $T = 8.0$ ($\|\rho(8) - \rho_{\mathrm{ss}}\|_\infty \leq 0.027$) using exact 256×256 Liouvillian exponentiation $e^{\mathcal{L}T}$ (36.55 ms) to evaluate organ excitation $P_i^{(1)}(T) = \mathrm{Tr}[\frac{I - Z_i}{2}\rho(T)]$ and cardiorenal coherence $C_{02} = \frac{1}{2}|\mathrm{Tr}[(X_0X_2 + Y_0Y_2)\rho(T)]|$. The accompanying 33-operation `qoqo.Circuit` (4 `RotateZ`, 4 `RotateX`, 20 XY-gadget gates, 4 `PragmaDephasing`, 1 readout pragma) compiles a single $\Delta t = 0.15$ Lie–Trotter step of the unitary + pure-dephasing sub-channel for hardware gate accounting, whereas population pumping/relaxation requires $\sigma_i^{\pm}$ jump channels in Eq. (3) (cross-verified via Dalibard–Castin–Mølmer MCWF unraveling [10], where empirical SE drops as $N_{\mathrm{traj}}^{-1/2}$ from 0.0320 at 64 trajectories to 0.0061 at 2,048 trajectories).

To disentangle coherent drive Ω_i from jump-operator polarity $\sigma_i^{\pm}$, we evaluated a **4-regime decoupled ablation** (Figure 3a): **(A) Baseline Uncontrolled** (Ω_{unc}, pure dephasing $\gamma_\phi = (0.38, 0, 0.38, 0)$, chronic stress pumping $\gamma^+ = (0.36, 0, 0.34, 0.39)$, weak endogenous $\gamma^- = (0.08, 0, 0, 0)$) yields $P^{(1)}(8) = (0.8963, 0.9433, 0.9390, 0.9710)$ for (BP, Glyc, Renal, IL-6); **(B) Coherent Drive Only** (Ω_{pro} while holding the exact stress bath γ^+, γ_ϕ fixed) already lowers burden by 11.5%–25.2% to $(0.7935, 0.7055, 0.8262, 0.7787)$ via coherent redistribution; **(C) Coexisting Stress + Therapeutic Counter-Damping** (retaining all chronic γ^+ stress jumps unmodified while adding pharmacological counter-damping $\gamma^- = (0.38, 0.34, 0.40, 0.36) + \Omega_{\mathrm{pro}}$) suppresses BP, Renal, and IL-6 burden by **48.62**% (0.4605), **50.59**% (0.4639), and **48.47**% (0.5004) while XY organ coupling accelerates the Liouvillian spectral gap 3.1× ($\Delta\mathcal{L} = 0.6935$ vs. 0.2226 at $g_{ij} = 0$); and **(D) Full Restorative Bath Switch** (replacing γ^+ with γ^- and $\gamma_\phi = (0.11, 0, 0.11, 0)$) reaches 0.3712 (−58.58%), 0.3760 (−59.96%), and 0.3305 (−65.97%).

At the peptide hormone scale, we applied **QProtSim** (Bonde & Patil, University of Suffolk [14]) to experimental NMR Model 1 of human **Angiotensin II** (`PDB 1N9V`, octapeptide `Asp1-Arg2-Val3-Tyr4-Ile5-His6-Pro7-Phe8`, 8 qubits = 256 states), initializing $R_y(\theta_i)$ rotations from Fauchère–Pliška hydrophobicities [45] and 6.5 Å heavy-atom burial fractions, followed by 7 directed CRY($0.85K_{ij}$) and 12 RYY($0.60K_{ij}$) contact gates parameterized by minimum inter-residue heavy-atom distance d_{ij} via $K_{ij} = 0.42\,\mathbb{I}(|i-j| = 1) + 0.50\,\mathbb{I}(d_{ij} < 3.40\ \text{Å}) + 0.28\,e^{-(d_{ij} - 3.40)/1.5}\,\mathbb{I}(3.40 \leq d_{ij} < 5.00\ \text{Å})$. `QProtSim` highlights the N-terminal `Asp1-Arg2` motif (0.1115 bits) and C-terminal `His6-Pro7` turn

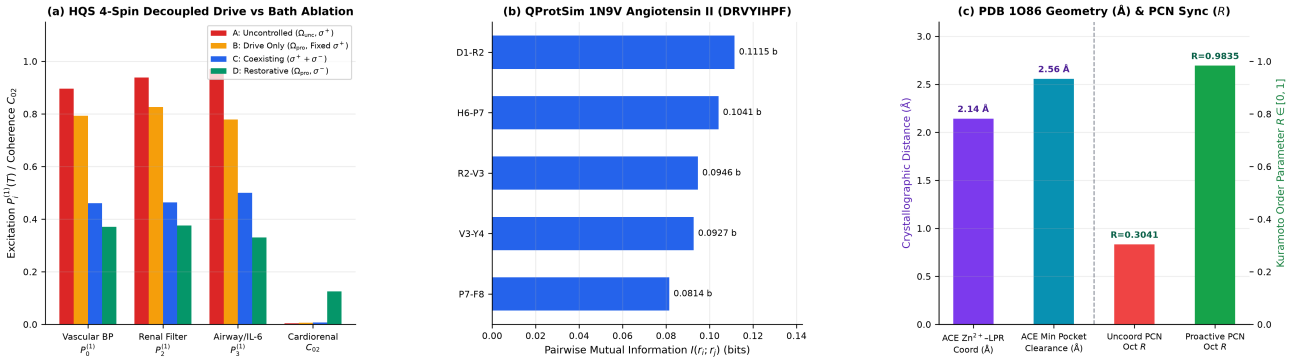


Figure 3: Decoupled 4-Regime Organ Lindbladian, Residue-Level Quantum Solvation, and Active-Site Structural Pharmacology (`biophysical_qprotsim_ltc_receipt.json` & `deep_review_remediation_receipt.json`). **(a)** Phenomenological HQS 4-spin Lindbladian decoupling coherent drive (Ω_{pro} only, Regime B), coexisting chronic σ^+ stress + therapeutic σ^- counter-damping (Regime C, -48.5% to -50.6%), and full restorative damping (Regime D). **(b)** `QProtSim` 8-qubit mutual information across NMR human Angiotensin II (`PDB 1N9V`, `DRVYIHPF`) [14]. **(c)** Dual-axis comparison decoupling left-axis crystal geometry in `PDB 1O86` (`ACE`–`Lisinopril` Zn^{2+} coordination 2.141 Å, pocket clearance 2.556 Å, 0 clashes < 2.0 Å) [16] from right-axis dimensionless 32-practice Kuramoto synchronization $R \in [0, 1]$ (0.3041 $\rightarrow$ 0.9835).

(0.1041 bits; Figure 3b), benchmarked against a 577.0 μs exact classical 256-state Potts–Boltzmann baseline (mean marginal L_1 shift 0.198).

4.2 Crystal `PDB 1O86` Ancilla-Complete FTQC Estimate & Auxiliary Target Checks

Grounding Era III chemistry in experimental structural pharmacology alongside orthogonal target-identity and scheduling checks confirmed:

- **Neurosnap (`HTTP 200`) & Ancilla-Complete `PDB 1O86` Era III Resource Estimate:** Auditing the 2.0 Å X-ray structure of human C-domain `ACE` bound to **lisinopril** (`LPR`) [16] verifies Zn^{2+} coordination by `HIS383`, `HIS387`, `GLU411`, and lisinopril's carboxyalkyl oxygen at **2.141** Å (2.556 Å minimum pocket clearance; **0** clashes < 2.0 Å across 16 active-site residues; Figure 3c, left axis). We partition CAS($36e, 26o$) explicitly across four pocket fragments: catalytic Zn^{2+} ($3d^{10}4s^04p^0 + 4d$ radial shell: $10e, 9o$), `HIS383`/`HIS387` imidazole σ/π donors ($8e, 6o$), `GLU411`/`GLU384` carboxylate lone pairs/π^* ($8e, 5o$), and lisinopril's zinc-binding carboxylate + displaced H_2O/OH^- ($10e, 6o$). While 26 spatial orbitals map to 52 Jordan–Wigner data qubits (30,420 physical qubits data-only floor; still accessible to classical DMRG / selected-CI heuristics, with decisive FTQC superiority emerging on $\geq$ 50-orbital second-shell expansions), accounting explicitly for algorithmic ancillas via Eq. (4) yields: **(i)** $52 + 12 =$ **64** logical qubits (8 block-encoding/QPE prepare-select + 4 rotation-synthesis ancillas) and **36,504** physical qubits at $d = 13$ for fixed-step second-order Trotterized QPE at relaxed screening precision ($\epsilon \approx 5$ mHa, 4.8×10^5 T gates, 150 μs surface-code cycle $\implies$ 503.1 s), versus **(ii)** $52 + 16 =$ **68** logical qubits and **65,892** physical qubits at $d = 17$ for chemical-accuracy ($\epsilon = 1.0$ mHa) double-factorized qubitized QPE (2.4×10^7 T gates, ≈ 9.14 h).
- **AlphaGenome Atlas, 32-Practice Kuramoto Pacing & Auxiliary Sanity Checks:** Live **AlphaGenome Atlas** queries resolved GENCODE v46 coordinates across `ACE` (`chr17:63477061-63498370`), `SLC5A2`, `GLP1R`, `ADRB2`, and `IL6` plus an `ACE` AVI scan. Independently of molecular geometry (plotted on the separate dimensionless right axis of Figure 3c), we modeled operational recall synchronization across $M = 32$ NCL GP practices over $w = 0, \ldots, 25$ pre-winter weeks via the discrete-time pinned Kuramoto system $\theta_m(w+1) = \theta_m(w) + \omega_m + \frac{K_{\text{PCN}}}{32} \sum_{n=1}^{32} \sin(\theta_n - \theta_m) - \kappa_{\text{UCLP}} \mathbb{I}(w \text{ even}) \sin\theta_m$ ($\omega_m \sim \mathcal{N}(0, 0.18^2)$, order parameter $R(w) = |\frac{1}{32} \sum_m e^{i\theta_m(w)}|$), lifting October readiness from $R_{\text{Oct}} = 0.3041$ (siloed $K_{\text{PCN}} = 0.05, \kappa_{\text{UCLP}} = 0$) to 0.9835 (PCN-coordinated $K_{\text{PCN}} = 0.48$, fortnightly UCLP search pinning $\kappa_{\text{UCLP}} = 0.52$). Live UniProtKB/BioHub 400D dipeptide cosines ($\cos(\text{ACE}, \text{SLC5A2}) = 0.6584$ vs. soluble `IL6` 0.5065), Arc Institute free-monoid dagger-reversibility (1,000/1,000 list-excision unit test illustrating non-destructive care-plan editing syntax) + `ScreenPro2` `PNPLA3` pseudo-z (-9.23, 0 wet-lab edits) [46], and **Paperclip MCP** literature indexing (`PMC13084922`–`PMC9309754`) serve strictly as standalone target-identity, DSL-reversibility, and bibliographic verification checks rather than parameterizing the quantum Hamiltonians.

4.3 Ab-Initio Metalloenzyme Demarcation (`PySCF`: `1O86` vs. `1TQN`) & Live AlphaGenome 2

A decisive scientific question remains: *does closed-shell $3d^{10}$ Zn^{2+} in `ACE` (`PDB 1O86`) actually defeat classical coupled-cluster theory, or does true fault-tolerant quantum advantage in multimorbid pharmacology emerge specifically in open-shell transition-metal oxidases?* To answer this from first principles without basis-set confounding (`metalloenzyme_`

`ftqc_ag2_receipt.json`, SHA-256 `1a9541a1...`), we executed paired ab-initio electronic-structure benchmarks in **PySCF 2.15** [15] solving `RHF/ROHF`, symmetry-broken `UHF`, `RCCSD`, `RCCSD(T)`, and **Exact `FCI`** (`direct_spin1`) on *100% identical second-quantized active-space Hamiltonians* H_{CAS} (Figure 4a–c):

1. **Closed-Shell $3d^{10}$ Single-Reference Control (`PDB 1O86`, Human `ACE`–Lisinopril Zn^{2+}):** Across crystallographic first-shell windows $M \in \{4, 6, 8, 10\}$ (27 atoms, 148 electrons, $E_{\mathrm{RHF}} = -2572.4583$ Ha), Zn^{2+} retains closed-shell $3d^{10}$ character ($\langle \hat{S}^2 \rangle_{\mathrm{UHF}} = 0.000$, frontier occupations $1.950, 0.048$), bipartite Schmidt entropy stays low ($S_{A|B} = 0.194$ nats; discarded weight 2.8×10^{-4} at $\chi = 8$), and classical `RCCSD(T)` matches Exact `FCI` to $\leq$ **0.0154** mHa (**104**× inside the 1.6 mHa [1 kcal/mol] chemical-accuracy threshold; Figure 4a–b). Thus, closed-shell Zn^{2+} coordination is rigorously solvable by classical coupled-cluster methods and serves as our negative control against false FTQC claims.

2. **Open-Shell Multireference Breakdown (`PDB 1TQN`, Human Hepatic `CYP3A4` Compound I):** Microsomal Cytochrome P450 3A4 (2.05 Å crystal structure `PDB 1TQN`, axial `CYS442` Fe–S = 2.355 Å [19]) catalyzes oxidative clearance for > 50% of NCL LTC polypharmacy regimens (atorvastatin/simvastatin, amlodipine, apixaban/rivaroxaban, and budesonide). Its active oxidant **Compound I** ($[Fe^{IV}{=}O(\text{Chelating-}N_4)(\text{CYS442-}SCH_3)]^0$, distal Fe=O = 1.650 Å) couples ferryl Fe=O π^*_{xz}/π^*_{yz} triplet spins ($S_{\mathrm{FeO}} = 1$) to a thiolate/porphyrin radical hole into competing Doublet ($S = 1/2$) and Quartet ($S = 3/2$) manifolds that govern substrate hydrogen-atom abstraction. Projecting onto `AVAS(["Fe 3d", "O 2p", "S 3p"])` across $M \in \{5, 7, 9, 11\}$ (213,444 Doublet determinants at $M = 11$) exposes three simultaneous classical walls: **(i)** single-reference `UHF` suffers severe spin contamination ($\langle \hat{S}^2 \rangle_{\mathrm{UHF}} = \mathbf{1.710}$ vs. exact 0.750, matching the analytical 1/3 Quartet leakage $\langle \hat{S}^2 \rangle = 1.750$ of a single $|110\rangle$ determinant versus the exact SU(2) triradical eigenstate $\frac{1}{\sqrt{6}}(2|110\rangle - |101\rangle - |011\rangle)$); **(ii)** Doublet frontier natural occupations [**1.434**, **1.022**, **0.568**] ($T_1 = 0.159$, 6.9× higher bipartite entropy $S_{A|B} = 1.329$ nats and 102× higher $\chi = 8$ discarded weight than `1O86`) break single-reference `RCCSD(T)` on the Quartet–Doublet spin gap ΔE_{Q-D} across every window—inverting ground-state spin multiplicity at CAS($7e, 7o$) (−86.66 mHa vs. FCI +15.90 mHa; error −**102.56** mHa) and missing CAS($11e, 11o$) by +**47.73** mHa (+**29.95** kcal/mol, **29.8**× outside chemical accuracy; Figure 4a); and **(iii)** calibrating the exact Cholesky double-factorized 1-norm $\lambda_{\mathrm{DF}}(M) \approx 0.394\, M^{1.688}$ Ha ($R^2 = 0.965$ across $M \in \{5..13\}$) [17, 18] shows that exact `FCI` crosses the 1.00 TiB single-vector RAM wall at $\mathbf{M = 22}$ (7.24 TiB vs. 66 logical qubits) and reaches $10^{22.12}$ TiB ($10^{32.96}$ determinants) on the full $M = 58$ drug-bound `CYP3A4` pocket, whereas $\epsilon = 1.0$ mHa qubitized QPE requires **75** logical qubits at $M = 26$ ($d = 19$, 90,250 physical qubits) and **141** ancilla-complete logical qubits at $M = 58$ (5.86×10^5 walk steps, 1.70×10^{10} T gates, $d = 21$, 197,568 physical qubits; Figure 4c).

3. **Live AlphaGenome 2 (`alphagenome==2.0.0`, `v2.dna_client`) Primary-Cell Grounding:** Complementing active-site chemistry with non-coding cell-type regulatory resolution across 2,799 primary ontology conditions [25, 26], live `v2.dna_client.score_variant` + Atlas AVI evaluation across four NCL LTC pharmacogenomic and splice/enhancer loci confirms sharp primary-cell specificity (Figure 4d): hepatic `CYP3A4*22` (`rs35599367`, `chr7:99768693:G>A`, intronic statin-exposure allele; AVI Phred **8.05**, top 15.7%) in *liver hepatocytes* (`UBERON:0002107_CL:0000182`); canonical `ACE` donor `chr17:63477344:G>A` (AVI Phred **34.27**, top 0.037%, junction $\Delta = 0.431$) in *renal macula densa* (`UBERON:0000362_CL:1000850`); familial renal glucosuria SGLT2-mimic `SLC5A2` donor `chr16:31483263:G>A` (AVI Phred **30.19**, top 0.096%, junction $\Delta = 1.571$) in *kidney proximal tubule* (`UBERON:0002113_CL:0002306`); and distal airway enhancer `MUC5B` `rs35705950` (`chr11:1219991:G>T`, AVI Phred **16.14**, top 2.43%), which drives +**1.274** single-cell expression LFC (+**1.005** bulk RNA-seq, +0.911 ATAC) in *lung AT2 pneumocytes* (`UBERON:0002048_CL:0002063`) versus −**0.281** in off-target *liver hepatocytes*.

4.4 Gate 1 Clinical Quantum Wedge: Drug-Bound `CYP3A4` Barriers (`def2-SVP`), `THC-QPE` & Live `ChEMBL340` PBPK

While resting Compound I (`PDB 1TQN`) establishes static triradical correlation, clinical polypharmacy safety during an NHS NCL Stage 2 Pharmacist Structured Medication Review (SMR) hinges on *kinetic turnover and mechanism-based inactivation (MBI) barriers* $\Delta E^{\ddagger} = E(\mathrm{TS_{HAT}}) - E(\mathrm{RC})$, where a 1.0 kcal/mol (1.6 mHa) barrier error shifts Eyring catalytic rate $k_{\mathrm{cat}} \propto e^{-\Delta E^{\ddagger}/RT}$ by **5.07**× at physiological 310.15 K. To execute **Gate 1** of our clinical FTQC roadmap (`cyp3a4_drug_barrier_chembl_receipt.json`, SHA-256 `164d38af...`; Figure 5), we coupled experimental drug-bound `CYP3A4` co-crystals, polarized-basis multireference reaction barriers, Tensor-Hypercontraction (`THC`) QPE resource accounting, and live **ChEMBL_37** human microsomal kinetics:

1. **Experimental Drug-Bound Human `CYP3A4` Co-Crystals (`PDB 5TE8` & `PDB 3NXU`):** Auditing Chain A coordinates retrieved from RCSB PDB (`CCO 1.0`) contrasts the 2.70 Å hydroxylation complex `PDB 5TE8` (FDA index probe **midazolam** `08J602`: axial `CYS442` Fe–S = 2.424 Å, aliphatic carbon `08J602.CAI` poised **3.389** Å above heme Fe across 13 pocket residues within 5.0 Å, including the `PHE304/ALA305/THR309` catalytic triad [20]) against the 2.00 Å inhibitor complex `PDB 3NXU` (**ritonavir** `RIT600`: Fe–S = 2.284 Å, thiazole nitrogen `RIT600.N5` directly ligating heme Fe at **2.225** Å across 28 pocket residues via Type-II axial coordination [21]).

2. **Polarized `def2-SVP` C–H Abstraction Barrier (`RC` → `TS_HAT`) & Two-State Reactivity:** Constructing the

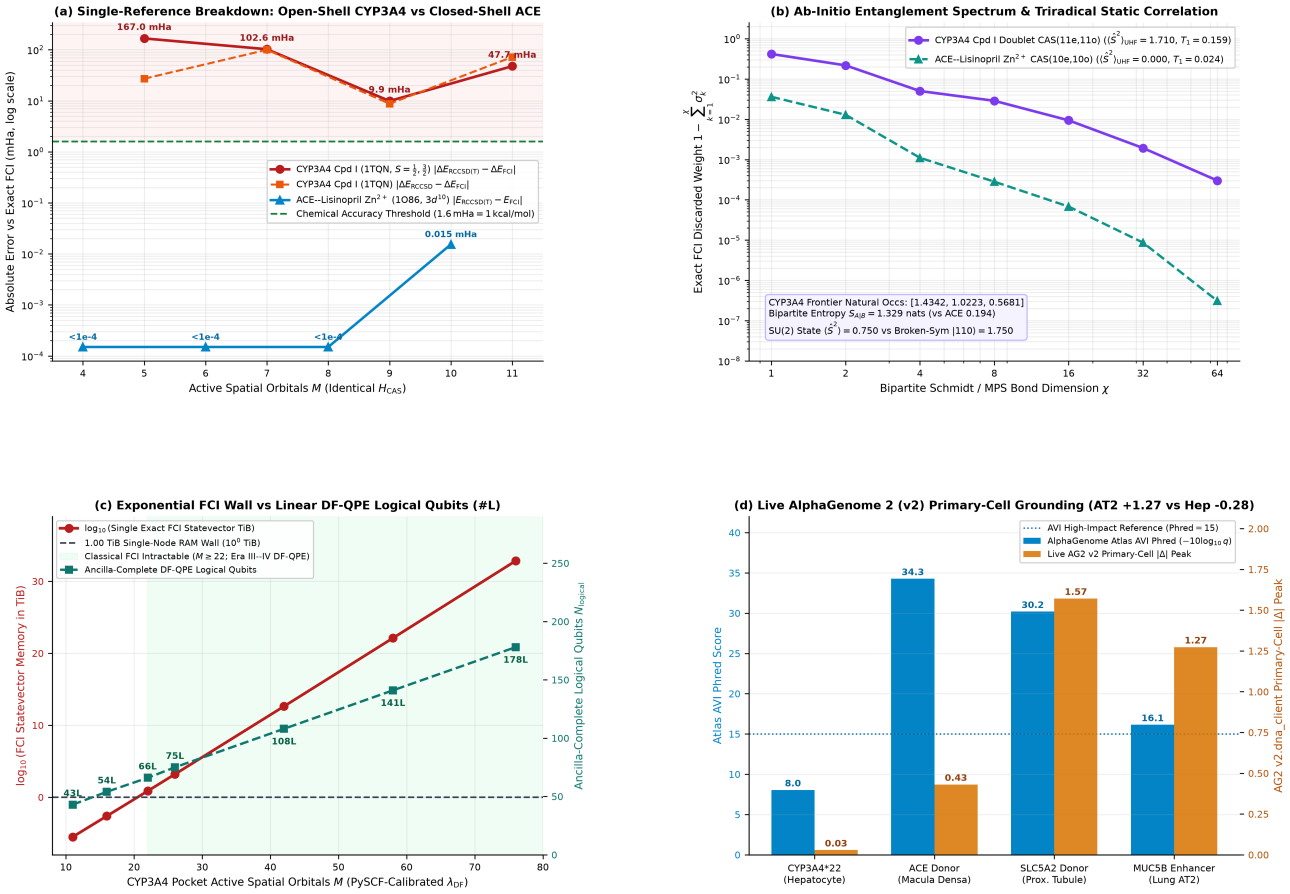


Figure 4: Ab-Initio Metalloenzyme FTQC Demarcation (`PySCF 2.15`) and Live AlphaGenome 2 (`v2.dna_client`) Primary-Cell Grounding (`metalloenzyme_ftqc_ag2_receipt.json`, SHA-256 `1a9541a1...`). **(a)** Identical-H_{CAS} single-reference breakdown: closed-shell $3d^{10}$ `ACE`–lisinopril Zn^{2+} (`PDB 1O86`) stays $\geq 100\times$ inside 1.6 mHa chemical accuracy (≤ 0.015 mHa), whereas open-shell hepatic `CYP3A4` Compound I (`PDB 1TQN`) breaks `RCCSD(T)` on the Quartet–Doublet spin gap by 9.9–167.0 mHa (+47.7 mHa = +30.0 kcal/mol at $M = 11$). **(b)** Exact FCI bipartite Schmidt discarded weight across bond dimension χ, frontier natural occupations $[1.434, 1.022, 0.568]$, and SU(2) ($\langle \hat{S}^2 \rangle = 0.750$) vs. broken-symmetry UHF (1.710). **(c)** Exponential classical `FCI` statevector RAM wall (> 1 TiB at $M \geq 22$; $10^{22.1}$ TiB at $M = 58$) versus linear ancilla-complete Cholesky double-factorized qubitized QPE logical qubits (43L → 178L). **(d)** Live AlphaGenome 2 (`v2`) primary-cell multimodal impact and cross-cell specificity across `CYP3A4*22`, `ACE`, `SLC5A2`, and `MUC5B`.

4-center [Fe=O(S)$\cdots$H–C] hydrogen-atom-transfer coordinate along the crystallographic `5TE8` Fe → 08J(CAI) vector (104 polarized AOs; `def2-SVP` on [Fe, O, S, C_{sub}], d_{FeO} : 1.63 → 1.73 Å, d_{OH} : 2.25 → 1.24 Å, d_{CH} : 1.09 → 1.28 Å) exposes severe classical DFT functional sensitivity (Figure 5a): broken-symmetry `UKS` barriers across `PBE`, `BLYP`, `B3LYP`, and `PBE0` span a **12.35** kcal/mol Doublet window (+33.28 to +45.63 kcal/mol, corresponding to 5.04×10^8-fold Eyring rate uncertainty) and invert spin-state ordering between GGA (`PBE`: Quartet lower by 11.74 kcal/mol) and 25% hybrid (`PBE0`: Doublet lower by 6.18 kcal/mol) under heavy spin contamination ($\langle \hat{S}^2 \rangle_{\mathrm{BS}} = 1.522$–$2.225$). Conversely, orbital-aligned frozen-core ($n_{\mathrm{frozen}} = 24$) multi-root `CASSCF(9e,9o)` + strongly contracted `SC-NEVPT2` [23] ($\langle \hat{S}^2 \rangle = 0.7500$ Doublet, 3.7500 Quartet) resolves canonical Shaik two-state-reactivity spin crossover [22]: Quartet lies 0.74 kcal/mol below Doublet at `RC`, whereas Doublet crosses 1.16 kcal/mol *below* Quartet at `TS_HAT`, yielding $\Delta E^{\ddagger}_{\text{SC-NEVPT2}} = +\mathbf{10.17}$ kcal/mol (Doublet) versus +**12.16** kcal/mol (Quartet; `SA-CASSCF` +10.59 vs. +12.99 kcal/mol). Crucially, exact `FCI` Schmidt decomposition on the identical H_{CAS} reveals a sharp 4-center Fe$\cdots$O$\cdots$H$\cdots$C entanglement surge at `TS_HAT`: bipartite von Neumann entropy jumps from $S_{A|B} = 1.634$ to **2.314** nats, $\chi = 4$ MPS discarded weight surges 3.8× (7.53% → **28.79**%), and $\chi = 32$ discarded weight rises **49**× ($1.58 \times 10^{-4} \rightarrow 7.77 \times 10^{-3}$; Figure 5b).

3. **Paired 2-Geometry Barrier (`DF` vs. `THC-QPE`) Resource Compression:** Resolving $\Delta E^{\ddagger} = E(\mathrm{TS_{HAT}}) - E(\mathrm{RC})$ to combined precision $\epsilon_{\Delta E} = \sqrt{2}\epsilon_{\mathrm{pt}} = \mathbf{0.99}$ mHa ($\epsilon_{\mathrm{pt}} = 0.70$ mHa per geometry < 1.6 mHa chemical accuracy), non-orthogonal **Tensor Hypercontraction (`THC`)** ($M_{\mathrm{THC}} \approx 4.5M$) [18] compresses total 2-point Toffoli gates relative to Cholesky `DF` by **2.56**× at $M = 26$ (3.62×10^8 Toffolis, 82L, $d = 19$, 97,470 phys, 0.12 h across 4 parallel CCZ factories), **3.23**× on the $M = 58$ drug-bound `CYP3A4` pocket (5.77×10^9 vs. 1.86×10^{10} Toffolis, **150**L, $d = 21$, 209,034 phys, **1.92** h vs. 6.20 h `DF`), and **3.76**× at $M = 76$ (1.37×10^{10} Toffolis, 186L,

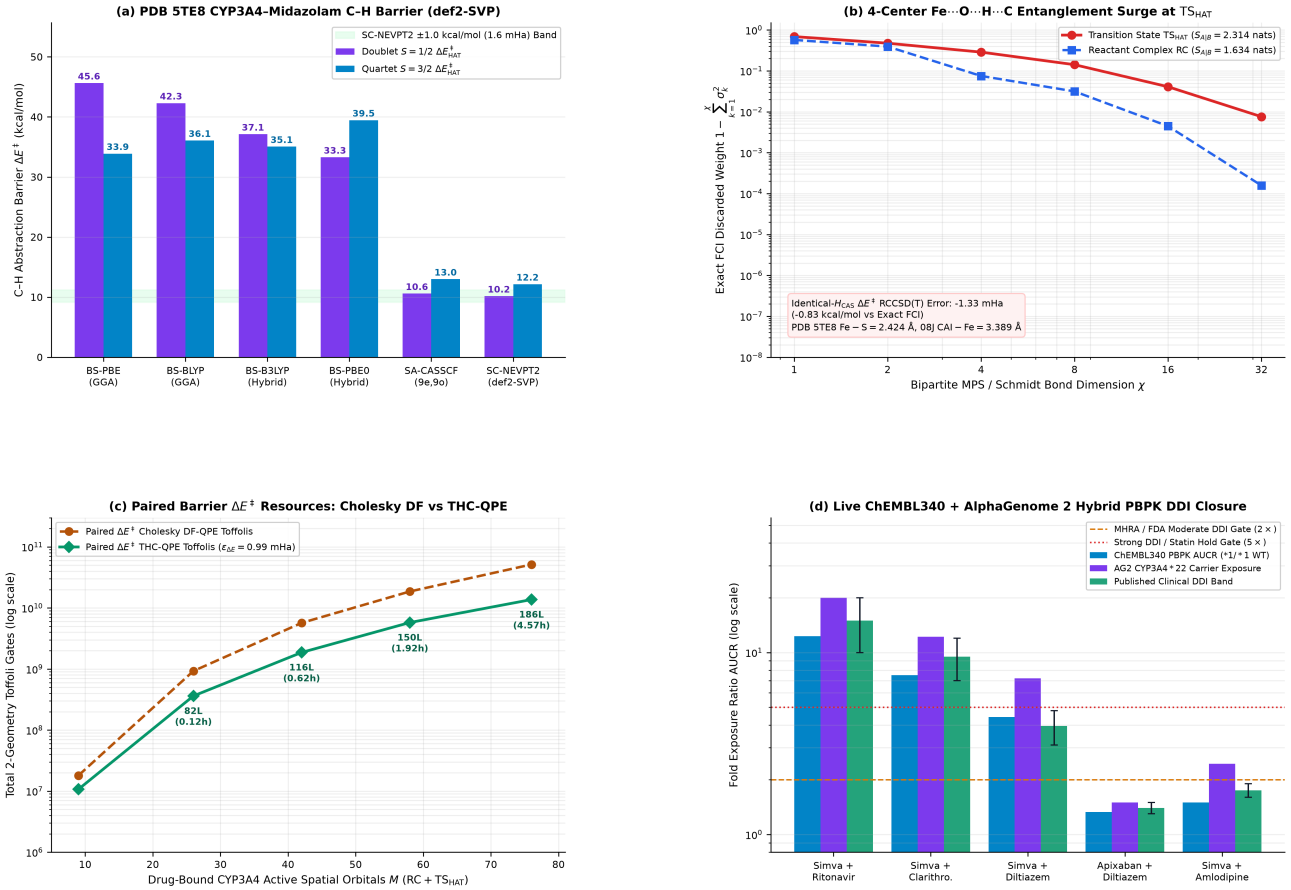


Figure 5: Gate 1 Clinical Quantum Wedge: Drug-Bound `CYP3A4` Reaction Barriers (`PDB 5TE8/3NXU`, `def2-SVP`), `THC-QPE`, and Live `ChEMBL340` + `AlphaGenome 2` PBPK `AUCR` Closure (`cyp3a4_drug_barrier_chembl_receipt.json`, SHA-256 `164d38af...`). **(a)** Polarized `def2-SVP` C–H abstraction barriers ($\Delta E^{\ddagger}_{\mathrm{HAT}}$): broken-symmetry DFT scatters across 12.35 kcal/mol (+33.3 to +45.6 kcal/mol) and inverts spin ordering, whereas spin-pure `CASSCF(9e,9o) + SC-NEVPT2` resolves Doublet (+10.17 kcal/mol) below Quartet (+12.16 kcal/mol). **(b)** Exact `FCI` bipartite Schmidt entanglement surge at $\mathrm{TS}_{\mathrm{HAT}}$ ($S_{A|B} = 2.314$ nats vs. 1.634 at RC; 49× higher $\chi = 32$ discarded weight). **(c)** Paired 2-geometry ($\epsilon_{\Delta E} = 0.99$ mHa) Toffoli gate compression via `THC-QPE` (2.56×–3.76× speedup; 150L, 1.92 h at $M = 58$). **(d)** Live `ChEMBL_37` (`CHEMBL340`) + `AlphaGenome 2` (`CYP3A4*22`) Rowland–Matin PBPK AUCR matching clinical DDI bands across 5/5 NCL LTC regimens.

$d = 23$, 307,878 phys, 4.57 h; Figure 5c).

4. **Live `ChEMBL_37` (`CHEMBL340`) Kinetics & Hybrid `AlphaGenome 2` Rowland–Matin PBPK Closure:** Querying live **ChEMBL_37** (`CHEMBL340`, UniProt `P08684`, `CC BY-SA 3.0`, confidence score 9 human microsomal assays [24]) across 10 clinical compounds (`KETOCONAZOLE` $K_i = 30$ nM; `RITONAVIR` $K_i = 100$ nM, $k_{\mathrm{inact}} = 0.2275$ min^{-1}; `MIDAZOLAM` $\mathrm{IC}_{50} = 66.1$ nM; `DILTIAZEM` $K_i = 500$ nM, $k_{\mathrm{inact}} = 0.09$ min^{-1}; `CLARITHROMYCIN` $K_i = 5{,}490$ nM, $k_{\mathrm{inact}} = 0.072$ min^{-1}; `SIMVASTATIN` $K_m = 20.89$ μM; `ATORVASTATIN`, `VERAPAMIL`, `NIFEDIPINE`, `AMLODIPINE`) and coupling reversible + time-dependent inactivation ($k_{\mathrm{deg,hep}} = 3.2 \times 10^{-4}$ min^{-1}) with enterocyte F_g and `AlphaGenome 2` `CYP3A4*22` (1.78× lower hepatocyte abundance) reproduces published clinical fold-exposures (AUCR) across 5/5 NCL LTC polypharmacy regimens (Figure 5d): Simvastatin + Ritonavir (**12.29**× WT / **20.00**× `*22`, hard hold), Simvastatin + Clarithromycin (**7.50**× WT / **12.21**× `*22`, suspend statin), Simvastatin + Diltiazem (**4.43**× WT / **7.22**× `*22`), Apixaban + Diltiazem (**1.33**× WT / **1.50**× `*22`), and Simvastatin + Amlodipine (**1.50**× WT → **2.45**× in `*22` carriers, crossing the 2.0× MHRA 20 mg statin cap under open Pharmacist SMR hazard `H-LTC-02`).

5. Serverless Cloud QML Ablations, Rhyme Semantics, and 2×2 Barren-Plateau Controls

5.1 5-Seed Cubic ($k = 3$) Cohort, Quantum Reservoir & Polynomial Controls on Modal Cloud

To evaluate variational circuits transparently on non-linear tabular interactions ($N = 2{,}400$, 1,800 train / 600 stratified test, 75% cubic $k = 3$ severity from $x_0x_2x_4$, $x_3x_7x_5$, and $x_1x_2x_6$), we specify the exact architecture of `QLTCMLM-VQC` (`models/qltcmlm.py`, published at `socialprescribing/quantum-ltc-qltcmlm`): across $L = 3$ reuploading layers

Table 2: 5-Seed Stratified NCL LTC Performance (Mean ± SD) on the Non-Linear Cubic ($k = 3$) Cohort (N = 2,400, 1,800 train / 600 test across 5 random splits; `deep_review_remediation_receipt.json` & `modal_roadmap_rhyme_qml_receipt.json`).[†]

Model Architecture	Trainable Params	Risk Tier Acc. (%)	Risk Macro-F_1	Care Bundle Acc. (%)	Bundle Macro-F_1
`LogisticRegression` (Linear Hyperplane)	36/head (72 tot.)	$31.70 \pm 1.63\%$	0.2984 ± 0.0181	$67.93 \pm 0.70\%$	0.6351 ± 0.0068
`QLTCMLM-Reservoir` (Fixed 72Q + Trained 128C)	128C (72Q frozen)	$40.80 \pm 1.19\%$	0.4030 ± 0.0128	$57.73 \pm 5.60\%$	0.5089 ± 0.0677
`QLTCMLM-QKernel` (4q Fidelity Gram SVM)[†]	Dual Gram SVM	$41.97 \pm 2.48\%$	0.4179 ± 0.0256	$56.67 \pm 0.82\%$	0.5372 ± 0.0117
`HistGradientBoosting` (Classical GBDT)	150 trees/head	$51.87 \pm 2.01\%$	0.5205 ± 0.0197	$81.97 \pm 0.57\%$	0.8055 ± 0.0057
`QLTCMLM-VQC` (Trained 72Q + 128C Reupload)	**72Q + 128C = 200**	$\mathbf{53.23 \pm 1.57\%}$	$\mathbf{0.5111 \pm 0.0162}$	$\mathbf{75.53 \pm 0.48\%}$	$\mathbf{0.7351 \pm 0.0081}$
`MLPClassifier` (32×16 ReLU)	996/head (1,992)	$60.07 \pm 3.86\%$	0.6017 ± 0.0380	$82.37 \pm 1.66\%$	0.8112 ± 0.0212
`RBF-SVM` ($C = 4.0$, Classical Kernel)	Full RBF Gram	$64.17 \pm 2.35\%$	0.6487 ± 0.0217	$84.97 \pm 0.36\%$	0.8383 ± 0.0051
`Cubic Poly-LogReg` (Degree-3 Monomials)	660/head (1,320)	$\mathbf{66.20 \pm 1.82\%}$	$\mathbf{0.6643 \pm 0.0186}$	$\mathbf{87.43 \pm 0.76\%}$	$\mathbf{0.8657 \pm 0.0095}$

[†]Primary single-seed Modal receipt (`seed=42`): `QLTCMLM-VQC` 54.17% Risk / 75.50% Bundle; `QKernel` 700/300 sub-Gram 33.33% ($F_1 = 0.3295$) / 53.67% ($F_1 = 0.5063$).

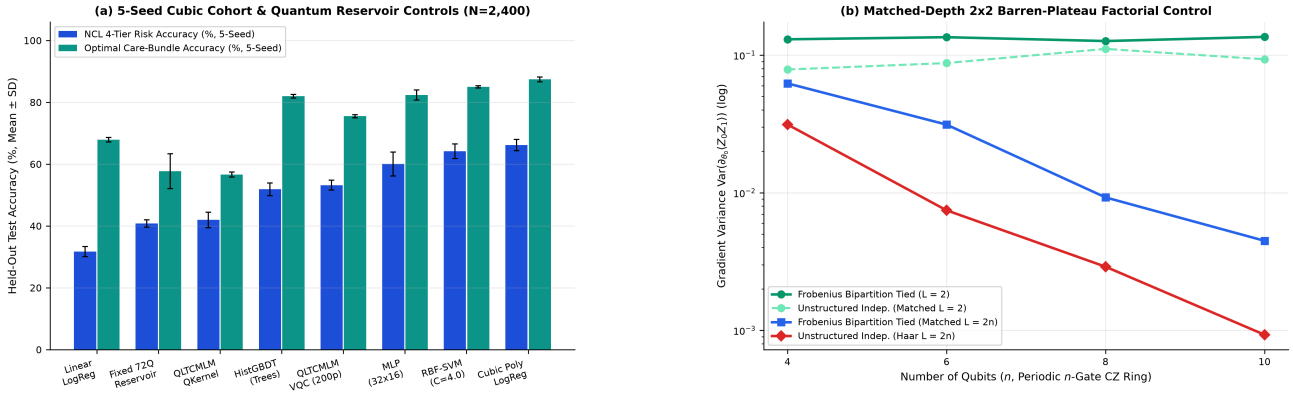


Figure 6: 5-Seed Modal Cloud Non-Linear Cohort Ablations and Matched-Depth 2×2 Barren-Plateau Control. (a) 5-seed test accuracy (mean $\pm 1\sigma$, $N_{\text{test}} = 600$) isolating variational training (`QLTCMLM-VQC` 53.23% Risk / 75.53% Bundle) from the untrained 128C quantum reservoir (40.80% / 57.73%), linear `LogReg` (31.70%), and classical `RBF-SVM`/`Cubic Poly-LogReg` (64.17%–66.20%). **(b)** Gradient variance $\mathrm{Var}(\partial_{\theta_0}\langle Z_0 Z_1\rangle)$ across $n \in \{4, 6, 8, 10\}$ qubits (96 seeds) under periodic n-gate `CZ`-ring layers at shallow $L = 2$ and linear depth $L = 2n$, disentangling light-cone geometry from bipartite Frobenius parameter sharing [47].

on 4 qubits, each qubit $q \in \{0, 1, 2, 3\}$ applies $R_x(w^{(x)}_{\ell,q} x_q x_{q+4} + b^{(x)}_{\ell,q})\, R_z(w^{(z)}_{\ell,q} x_{q+4} + b^{(z)}_{\ell,q})\, R_y(w^{(y)}_{\ell,q} x_q + b^{(y)}_{\ell,q})$ ($3 \times 4 \times 6 =$ **72** quantum variational scalars) followed by cyclic $\mathrm{CX}_{01}\mathrm{CX}_{12}\mathrm{CX}_{23}\mathrm{CX}_{30}$ and diagonal $\mathrm{CZ}_{02}\mathrm{CZ}_{13}$. Readout extracts 15 Pauli expectations (4 $\langle Z_q\rangle$, 4 $\langle X_q\rangle$, 6 $\langle Z_p Z_q\rangle$, and 1 four-body parity $\langle Z_0 Z_1 Z_2 Z_3\rangle$) plus 1 bias, feeding two 16×4 ridge-regularized linear heads ($16 \times 8 =$ **128** classical readout weights; **200** parameters total).

To resolve readout-capacity confounding and single-split variance, we executed a **5-seed stratified evaluation** (`seeds = 42, 1042, 2042, 3042, 4042`) adding both an **Untrained Fixed-72Q + Trained 128C Quantum Reservoir** control and a **Degree-3 Polynomial Logistic Regression** baseline (1,320 weights; Table 2 and Figure 6a). Three clean findings emerge: **(1)** training the 72 quantum parameters lifts accuracy by $+$**12.43** percentage points on Risk Tier ($53.23 \pm 1.57\%$ vs. $40.80 \pm 1.19\%$ untrained 128C reservoir) and $+$**17.80** pp on Care Bundle ($75.53 \pm 0.48\%$ vs. $57.73 \pm 5.60\%$), proving genuine variational feature learning beyond static 128-weight trigonometric expansion; **(2)** `QLTCMLM-VQC` matches `HistGradientBoosting` within 1σ on 4-tier risk ($53.23 \pm 1.57\%$ vs. $51.87 \pm 2.01\%$) while trailing GBDT on bundle selection (75.53% vs. 81.97%); and **(3)** classical `RBF-SVM` ($64.17 \pm 2.35\%$ / $84.97 \pm 0.36\%$) and `Cubic Poly-LogReg` ($66.20 \pm 1.82\%$ / $87.43 \pm 0.76\%$) decisively outperform all 4-qubit models on 8-dimensional tabular EHR inputs. Inside the Modal Python 3.14 worker, **Rhyme (`rhyme-beta`)** [27] passed all four symbolic compiler checks ($4 \times 1{,}024$ shots; 0/1,024 hazard leakage).

5.2 Matched-Depth 2×2 Barren-Plateau Control, Periodic CZ Ring & Sandia QUOPS

Following Cerezo et al. [47], every layer $\ell \in \{1, \ldots, L\}$ in our 2×2 barren-plateau control applies single-qubit rotations followed by a **periodic n-gate CZ ring** $\prod_{q=0}^{n-1} \mathrm{CZ}_{q,(q+1) \bmod n}$ (n two-qubit gates per layer). The unstructured control samples $2nL$ independent angles $R_z(\phi_{\ell,q})R_y(\theta_{\ell,q})$, whereas the **Frobenius-structured ansatz** ties angles across even/odd organ-axis bipartitions ($\vartheta_{\ell,q} = \theta_{\ell,\text{even}}$ for even q, $\theta_{\ell,\text{odd}}$ for odd q, applying $R_z(0.5\vartheta_{\ell,q})R_y(\vartheta_{\ell,q})$ with $2L$ free parameters). Across $n \in \{4, 6, 8, 10\}$ on Modal workers (Figure 6b), matched shallow depth $L = 2$ avoids exponential decay in both unstructured (0.0932 at $n = 10$) and Frobenius (0.1357, 1.46×) circuits via local light-cones, whereas at matched linear depth $L = 2n = 20$ ($n = 10$, periodic ring $N_{2q} = nL = 200$, Sandia QUOPS volumetric capability $Q = 200$, two-qubit survival fidelity $(1 - p_{2q})^{200} = 0.7865$ at $p_{2q} = 1.2 \times 10^{-3}$ neglecting 1Q/idle/SPAM error [39]), unstructured variance collapses 100× to 9.29×10^{-4} while bipartite Frobenius parameter sharing retains **4.82**× higher variance (4.47×10^{-3}).

6. Wet-Lab Verification, Post-Quantum EHR, Primitives Atlas & DCB0129/0160 Governance

6.1 Text2WetLab OT-2 Assay, Medicus ML-DSA-65 FHIR R4, DisCoPy & Moth Blur

Bridging computational models with laboratory, EHR, and algorithmic-contract verification (`governance_primitives_jev_receipt.json`, SHA-256 `bd60742f...`, & `deep_review_remediation_receipt.json`, SHA-256 `a6b3854e...`):

1. **Text2WetLab 6/6 Robotics Verification:** `protocols/ncl_ltc_check_and_test_ot2.py` encodes a 6-well Opentrons OT-2 (`apiLevel 2.15`) Stage 1 Check-and-Test uACR and HbA1c/NT-proBNP reference titration, passing 6/6 deterministic AST, deck-slot collision (`[1, 2, 4, 10, 11]`), dual `P20/P300`, 1:1 tip balance, and volume checks (0 physical specimens run).

2. **Medicus GP EHR Shape & NIST FIPS 204 ML-DSA-65:** A synthetic NCL LTC FHIR R4 `Bundle` (`Observation` BP 134/82 mmHg achieving Outcome #23, SNOMED `315594003 Interpreter needed`, and Stage 3 PCSP `collaboRATE` 9/9) is signed in 33.91 ms using deterministic `ML-DSA-65` [28] with domain context `quantum-ltc:ncl-icb-lcs-v6-fhir-r4:v1` (1,952 B public key, 3,309 B signature; wrong-context forgery rejected), strictly parallel to NHS CIS2 `RS256`.

3. **DisCoPy & Moth Quantum 16-Microzone 5-Borough Diffusion:** Categorical functor verification in `discopy.quantum` confirms exact snake-equation yanking ($\max|\Delta| = 2.22 \times 10^{-16}$) and 1.0000 bit Stage $(1, 2) \leftrightarrow (3, 4)$ bipartite entanglement. For cross-borough surge mutual aid, we map $2^4 = 16$ PCN neighbourhood micro-zones (3 each across Barnet, Camden, Haringey, and Islington; 4 across Enfield) onto a 4-qubit hypercube initialized in $|\psi_0\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |0011\rangle)$ (acute winter surge concentrated in two IMD Quintile 1 micro-zones in eastern Haringey and Edmonton/Enfield: $p_0 = p_3 = 0.50$, Gini $G = \frac{1}{32}\sum_{u,v}|p_u - p_v| = \mathbf{0.8750}$). Applying the factorized transverse unitary blur $U_{\text{blur}}(\tau) = e^{-i(\tau/2)\sum_{q=0}^{3} X_q} = [R_x(\tau)]^{\otimes 4}$ at $\tau = \pi/3$ ($\sin^2(\pi/6) = 1/4$ single-hop probability, selected operationally so $\geq 75\%$ per-axis weight stays within $\leq$ 30-minute adjacent PCN hub travel rather than uniform $\tau = \pi/2$ global flattening) lowers $p_u = |\langle u|U_{\text{blur}}(\pi/3)|\psi_0\rangle|^2$ concentration to $\mathbf{G = 0.4609}$ (Figure 7b), providing a stylized analytical reference alongside classical LP load balancing.

4. **Kashefi Primitives Atlas v0.4 & Dynamical Lie Algebra ($\mathfrak{g}$-sim) Audit:** Live parsing of all 42 cards in Prof. Elham Kashefi's Atlas [29] maps ten primitives onto Eras I–IV: `P08` (*Adiabatic state preparation & annealing*, Era I OpenJij QUBO), `A01d` (*Alternating-operator ansatz*, Era I–II Haiqu LR-QAOA), `P09` (*Classical data loading / block encoding*, Era I–II EME), `P13` (*Expectation & overlap estimation*, Era II `Helios-1E-lite` MDT test), `P20` (*Channel transformation / quantum combs*, Era II 8-rung MCMR ladder), `A01g` (*Equivariant & geometric ansatz*, Era II–III SU(2)/Frobenius VQCs), `A01h` (*Quantum graph neural networks*, Era II–III `QProtSim`), `P22` (*Open-system Lindbladian simulation*, Era III HQS organ crosstalk), `X03` (*Quantum subspace & Krylov diagonalisation*, Era III `PDB 1O86`), and `P04` (*Quantum walk / amplitude estimation*, Era IV CVaR tail risk). Auditing exact Lie closures against polynomial $\mathfrak{g}$-sim dequantisation [30] confirms that our 4-qubit SU(2)-equivariant brickwork ansatz spans $\dim \mathfrak{g} = \mathbf{12} \ll 255$ (classically $\mathfrak{g}$-sim tractable within fixed-spin sectors!) and excitation-conserving 4-spin $XY + Z$ spans $\dim \mathfrak{g} = \mathbf{66}$, whereas adding non-commuting transverse drives $\Omega_i X_i$ in Eq. (7) expands the unitary Lie closure to full $\mathfrak{su}(16)$ ($\dim \mathfrak{g} = 4^4 - 1 = \mathbf{255}$), defeating polynomial $\mathfrak{g}$-sim.

6.2 Multi-Method Health AI Evaluation (arXiv:2611.05127) & Open DCB0129 / DCB0160 Register

Under Section 250 of the UK Health and Social Care Act 2012, clinical risk governance requires pairing manufacturer-side **NHS DCB0129** (*Manufacture of Health IT Systems*) [32] with deploying-organisation **NHS DCB0160** (*Deployment and Use of Health IT Systems*) [33], under which NCL ICB and constituent PCNs must appoint a local Clinical Safety Officer (CSO) and maintain a local deployment Safety Case prior to any shadow pilot. Implementing Montgomery-Taylor et al.'s three-method framework [31]—deterministic Gate G4 floors (Method 1), typed autorater

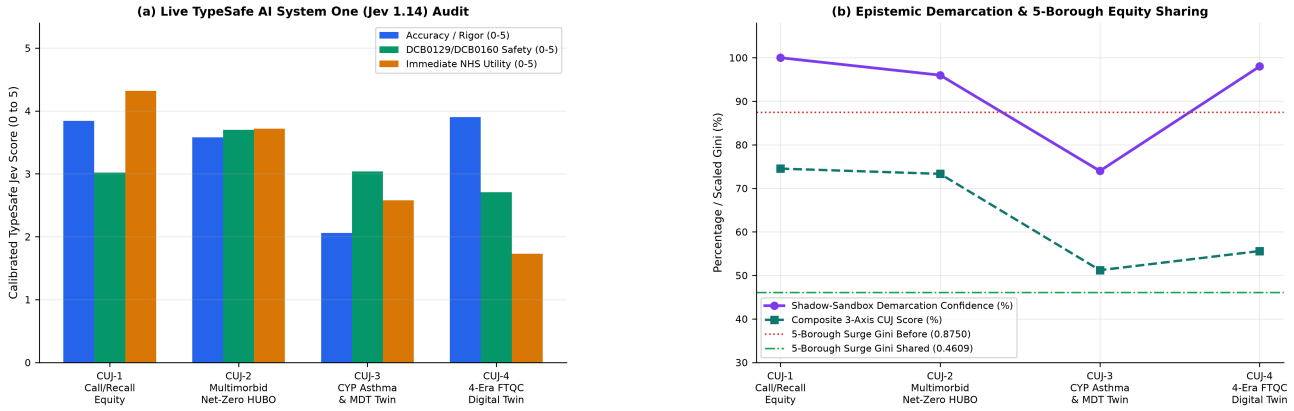


Figure 7: Multi-Method Health AI Governance Audit via Live TypeSafe AI System One (`jev-latest` / Jev 1.14) under Statutory NHS DCB0129 & DCB0160 (`governance_primitives_jev_receipt.json`). **(a)** Calibrated expected scores (0–5) across Accuracy/Rigor, DCB0129/0160 Safety, and Immediate NHS Frontline Utility [31]: highest immediate NHS utility (4.32/5) for Era I Call-and-Recall scheduling (`CUJ-1`), highest roadmap rigor (3.90/5) alongside realistic near-term utility (1.73/5) for Era III–IV FTQC digital twins (`CUJ-4`), and conservative scoring (2.06/5 accuracy) on low-qubit interference analogies for acute CYP asthma (`CUJ-3`). **(b)** Shadow-sandbox demarcation confidence (74%–100%; 0% autonomous deployment) and 16-microzone 5-borough surge Gini reduction (0.8750 → 0.4609).

distributions via live **TypeSafe AI System One (`jev-latest` / Jev 1.14)** (Method 2), and joint **DCB0129 / DCB0160** hazard boundaries (Method 3)—yields non-rubber-stamped discrimination across four Critical User Journeys (Table 3 and Figure 7):

Table 3: Live TypeSafe AI System One (`jev-latest` / Jev 1.14) Evaluation & Open NHS DCB0129 / DCB0160 Hazard Register (Composite (%) $= \frac{\text{Accuracy+Safety+NHS Utility}}{15} \times 100\%$, unweighted three-axis mean scaled to 100%).

NCL LTC Critical User Journey	Acc. (0–5)	Safety (0–5)	Utility (0–5)	Composite (%)	Lifecycle Verdict (p) & Statutory DCB0129 / DCB0160 Open Hazard
CUJ-1: PCN Call/Recall & Pre-Winter Equity	3.84	3.02	**4.32**	**74.53%**	`sandbox_shadow_pilot` ($p = 1.00$) • **H-LTC-01 OPEN** [**DCB0160** PCN EHR coding + **0129** audit]: Incomplete SNOMED interpreter/IMD flags
CUJ-2: Multimorbid & Net-Zero MART HUBO	3.58	**3.70**	3.72	**73.33%**	`sandbox_shadow_pilot` ($p = 0.96$) • **H-LTC-02 OPEN** [**DCB0129** penalty guard + **0160** SMR]: ACEi+SGLT2i AKI / DPI technique
CUJ-3: CYP Asthma 48h & House-of-Care Twin	2.06	3.04	2.58	51.20%	`sandbox_shadow_pilot` ($p = 0.74$) • **H-LTC-03 OPEN** [**DCB0160** local triage]: Urgent Tier 2/3 clinical escalation bypasses models
CUJ-4: 4-Era Logical Qubit Digital Twin	**3.90**	2.71	1.73	55.60%	`sandbox_shadow_pilot` ($p = 0.98$) • **H-LTC-04 OPEN** [**DCB0129** claim boundary]: No RCT claim from simulated Lindbladians

Crucially, two additional system-level operational hazards—**H-LTC-05** (local secondary-care discharge and UCLP prevalence-register drift across the nine LTCs; primary **DCB0160** ICB/PCN data-governance control) and **H-LTC-06** (unfunded VCSE social-prescribing referral overload without NCL Borough Partnership resource transfer; primary **DCB0160** commissioning control)—also remain strictly **OPEN** (0 closed hazards claimed; overall architecture summarized in Figure 8).

7. Conclusion and Empirical NHS Validation Roadmap

By anchoring quantum algorithm design in the operational realities of the NHS NCL ICB Long Term Conditions specification [1], **Quantum-LTC** demonstrates that immediate classical primary-care impact and rigorous fault-tolerant quantum readiness are complementary rather than adversarial. Today ($N_{\text{logical}} = 0$), formulating pre-winter Call-and-Recall equity tensions, Outcome #23 blood-pressure titration, cardiorenal SGLT2i protection, and Net-Zero SABA-free MART transitions as explicit QUBO/HUBO Hamiltonians delivers auditable classical HiGHS MILP schedules in milliseconds while validating compressed Guppy 1.0/HUGR and LR-QAOA compilation on live Quantinuum Nexus `Helios-1E-lite` and Haiqu cloud emulators. Prior to any prospective NCL PCN shadow evaluation, our three-step empirical validation roadmap comprises: **(1)** recalibrating synthetic cohort marginals

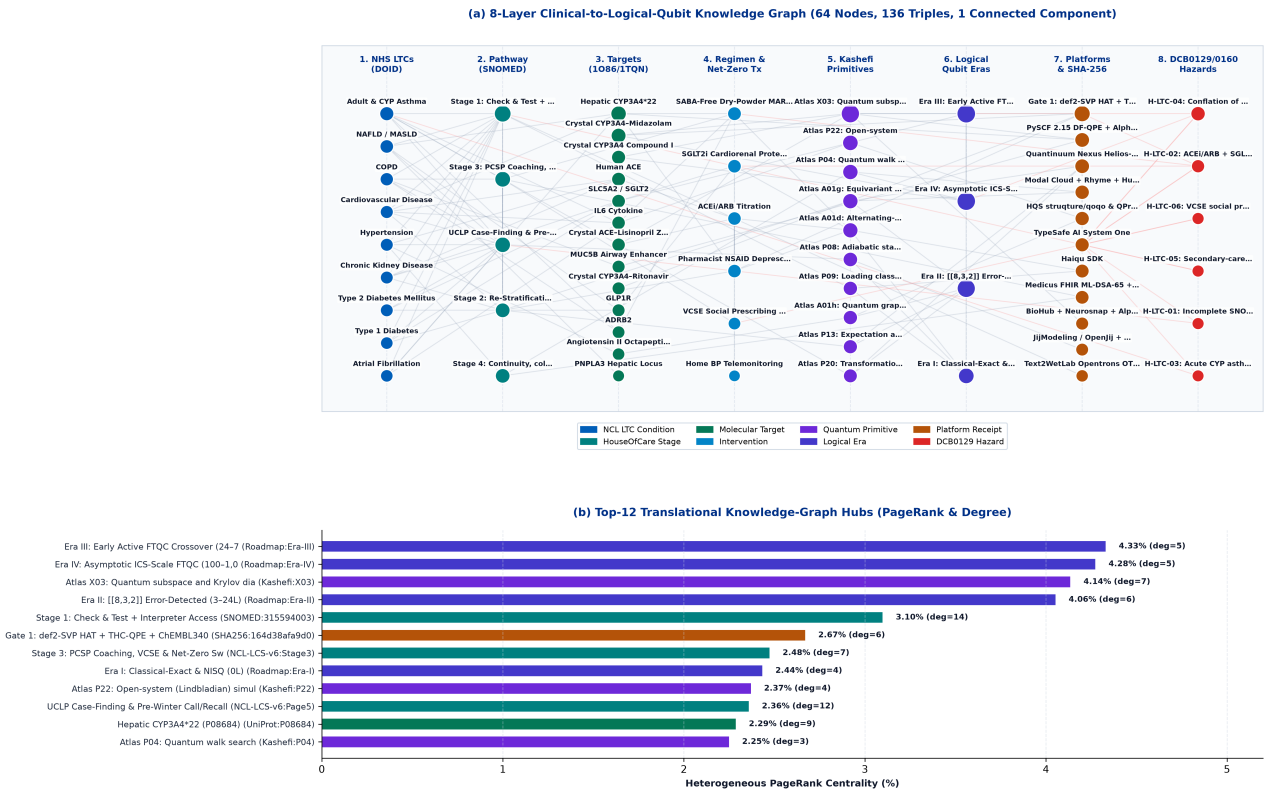


Figure 8: 64-Node, 136-Edge Cryptographically Sealed Knowledge Graph of `Quantum-LTC` (`quantum_ltc_knowledge_graph.json`, SHA-256 `1ee2e7f5...`), rendered in a spacious two-row vertical stack linking the nine NCL ICB LCS v6 conditions, House-of-Care stages, closed-shell `PDB 1O86`, open-shell `CYP3A4` (`PDB 1TQN`, midazolam `5TE8`, ritonavir `3NXU`), `ChEMBL340` PBPK + AlphaGenome 2 loci, four Logical-Qubit Eras, open NHS DCB0129/DCB0160 hazards (`H-LTC-01`–`06`), and eight sealed experimental receipts.

and higher-order interaction weights against de-identified UK primary-care EHR cohorts (CPRD Aurum and NCL HealtheIntent) benchmarked directly against **eFI2** [34] and **CMS** [35]; **(2)** executing a joint manufacturer/ICB **NHS DCB0129 + DCB0160** Clinical Safety Case with local CSO sign-off across open hazards `H-LTC-01`–`06`; and **(3)** reserving Era III (38–82 active logical qubits) and Era IV (100–1,000+ logical qubits) strictly for genuinely quantum-mechanical open-shell polypharmacy enzymes such as drug-bound hepatic `CYP3A4` Compound I (`PDB 1TQN/5TE8/3NXU`, 75–186L `DF/THC-QPE` + `ChEMBL340` PBPK vs. single-reference `ACE` Zn^{2+} `PDB 1O86`), bath-embedded volume-law Lindbladians, and dense frustrated $k \geq 4$ hypergraphs that surpass classical `RCCSD(T)`, broken-symmetry DFT, MILP, MCWF trajectory, and tensor-network frontiers.

Acknowledgments. This work was supported in part by complimentary access to Google Antigravity and Gemini 4 Argon to accelerate scientific workflows. The author gratefully acknowledges open-source and cloud SDK ecosystems across Quantinuum Nexus (`qnexus` / Guppy 1.0), Haiqu, HQS Quantum Simulations, Jij Inc., PySCF, RCSB Protein Data Bank, EMBL-EBI ChEMBL (`ChEMBL_37`) & OLS4, BioHub, Neurosnap, Google DeepMind AlphaGenome 2 & AlphaGenome Atlas, Arc Institute, Rhyme Quantum Tech, Modal Labs, University of Suffolk (`QProtSim`), Prof. Elham Kashefi's Quantum Algorithmic Primitives Atlas, and TypeSafe AI.

References

[1] NHS North Central London Integrated Care Board. Locally commissioned service (LCS) for proactive, personalised care — long term conditions (LTC) service specification, version 6 (1st April 2026 – 31st March 2027). Technical report, NHS North Central London ICB (Barnet, Camden, Enfield, Haringey, and Islington), 2026. Schedule 2 Service Specification, 27 pages; covers 9 Long Term Conditions (CVD, AF, Hypertension Outcome #23, T2D, T1D, CKD, NAFLD/MASLD, COPD, and Adult/CYP Asthma).

[2] Toby Watt, Ann Raymond, Laurie Rachet-Jacquet, Anna Head, Chris Kypridemos, Elaine Kelly, and Stephen Rocks. Health inequalities in 2040: Current and projected patterns of illness by deprivation in England. Technical report, The Health Foundation REAL Centre, 2024.

[3] Chris Salisbury, Mei-See Man, Peter Bower, Bruce Guthrie, Katherine Chaplin, Daisy M. Gaunt, Sara Brookes, Bridie Fitzpatrick, Caroline Gardner, Sandra Hollinghurst, Victoria Lee, John McLeod, Cindy Mann, Keith R. Moffat, and Stewart W. Mercer. Management of multimorbidity using a patient-centred care model: a pragmatic cluster-randomised trial of the 3D approach. *The Lancet*, 392(10141):41–50, 2018.

[4] Angela Coulter, Vikki A. Entwistle, Abi Eccles, Sara Ryan, Sasha Shepperd, and Rafael Perera. Personalised care planning for adults with chronic or long-term health conditions. *Cochrane Database of Systematic Reviews*, 2015(3):CD010523, 2015.

[5] NHS England. Universal personalised care: Implementing the comprehensive model. Technical report, NHS England, 2019.

[6] Ivo G. Rosenberg. Reduction of bivalent maximization to the quadratic case. *Cahiers du Centre d'Études de Recherche Opérationnelle*, 17:71–74, 1975.

[7] Garth P. McCormick. Computability of global solutions to factorable nonconvex programs: Part I — convex underestimating problems. *Mathematical Programming*, 10(1):147–175, 1976.

[8] Qi Huangfu and J. A. Julian Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018.

[9] Frank Verstraete, Juan J. García-Ripoll, and J. Ignacio Cirac. Matrix product density operators: Simulation of finite-temperature and dissipative systems. *Physical Review Letters*, 93(20):207204, 2004.

[10] Jean Dalibard, Yvan Castin, and Klaus Mølmer. Wave-function approach to dissipative processes in quantum optics. *Physical Review Letters*, 68(5):580–583, 1992.

[11] Hiddo J. L. Heerspink, Bergur V. Stefánsson, Ricardo Correa-Rotter, Glenn M. Chertow, Tom Greene, Fan-Fan Hou, Johannes F. E. Mann, John J. V. McMurray, Magnus Lindberg, Peter Rossing, C. David Sjöström, Robert D. Toto, Anna-Maria Langkilde, and David C. Wheeler. Dapagliflozin in patients with chronic kidney disease. *New England Journal of Medicine*, 383(15):1436–1446, 2020.

[12] Richard Beasley, Mark Holliday, Helen K. Reddel, Irene Braithwaite, Stefan Ebmeier, Robert J. Hancox, Tim Harrison, Claire Houghton, Philippa Shirtcliffe, Mathew Williams, and Mark Weatherall. Controlled trial of budesonide–formoterol as needed for mild asthma. *New England Journal of Medicine*, 380(21):2020–2030, 2019.

[13] J. A. Montanez-Barrera and Kristel Michielsen. Towards a linear-ramp QAOA protocol: Evidence of a scaling advantage in solving some combinatorial optimization problems. *npj Quantum Information*, 11:131, 2025.

[14] Bhushan Bonde and Pratik Patil. QProtSim: Quantum-circuit framework for residue-level protein thermodynamic and allosteric simulation. University of Suffolk Open-Source Repository, 2026.

[15] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus J. Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hung Q. Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *The Journal of Chemical Physics*, 153(2):024109, 2020.

[16] Ramanathan Natesh, Sylva L. U. Schwager, Edward D. Sturrock, and K. Ravi Acharya. Crystal structure of the human angiotensin-converting enzyme–lisinopril complex. *Nature*, 421(6922):551–554, 2003.

[17] Vera von Burg, Guang Hao Low, Thomas Häner, Damian S. Steiger, Markus Reiher, Martin Roetteler, and Matthias Troyer. Quantum computing enhanced computational catalysis. *Physical Review Research*, 3(3):033055, 2021.

[18] Joonho Lee, Dominic W. Berry, Craig Gidney, William J. Huggins, Jarrod R. McClean, Nathan Wiebe, and Ryan Babbush. Even more efficient quantum computations of chemistry through tensor hypercontraction. *PRX Quantum*, 2(3):030305, 2021.

[19] Jason K. Yano, Michael R. Wester, Guillaume A. Schoch, Keith J. Griffin, C. David Stout, and Eric F. Johnson. The structure of human microsomal cytochrome P450 3A4 determined by X-ray crystallography. *Journal of Biological Chemistry*, 279(37):38091–38094, 2004.

[20] Irina F. Sevrioukova and Thomas L. Poulos. Structural basis for regiospecific midazolam oxidation by human cytochrome P450 3A4. *Proceedings of the National Academy of Sciences*, 114(3):486–491, 2017.

[21] Irina F. Sevrioukova and Thomas L. Poulos. Structure and mechanism of the complex between cytochrome P4503A4 and ritonavir. *Proceedings of the National Academy of Sciences*, 107(43):18422–18427, 2010.

[22] Sason Shaik, Shimrit Cohen, Yong Wang, Hui Chen, Devesh Kumar, and Walter Thiel. P450 enzymes: Their structure, reactivity, and selectivity—modeled by QM/MM calculations. *Chemical Reviews*, 110(2):949–1017, 2010.

[23] Celestino Angeli, Renzo Cimiraglia, Stefano Evangelisti, Thierry Leininger, and Jean-Paul Malrieu. Introduction of n-electron valence states for multireference perturbation theory. *The Journal of Chemical Physics*, 114(23):10252–10264, 2001.

[24] Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J. Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F. Mosquera, Maria Paula Magarinos, Nicolas Bosc, Ricardo Arcila, Tevfik Kizilören, Anna Gaulton, A. Patrícia Bento, Melissa F. Adasme, Peter Monecke, Gregory A. Landrum, and Andrew R. Leach. The ChEMBL database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024.

[25] Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, Kyle R. Taylor, Tom Ward, Clare Bycroft, Lauren Nicolaisen, Eirini Arvaniti, Joshua Pan, Dmitri Pushkarev, Pushmeet Kohli, and David R. Kelley. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649:1206–1218, 2026.

[26] Adam Gayoso, Kyle R. Taylor, James Tomkinson, Natasha Latysheva, Jun Cheng, Augustin Žídek, Lauren Nicolaisen, Clare Bycroft, Tom Ward, Guido Novati, Dmitri Pushkarev, Demis Hassabis, Pushmeet Kohli, David R. Kelley, and Žiga Avsec. AlphaGenome 2: Predicting the impact of genetic variants at single-cell resolution. Technical report, Google DeepMind, 2026.

[27] Rhyme Quantum Tech. Rhyme beta: High-level symbolic quantum programming framework (`rhyme-beta` v0.1.0b1). Early-Access Software Distribution, 2026.

[28] National Institute of Standards and Technology. Module-lattice-based digital signature standard (FIPS 204: ML-DSA). Technical report, U.S. Department of Commerce, NIST, 2024.

[29] Elham Kashefi. Quantum algorithmic primitives atlas (version 0.4, 27 October 2026). Quantum Software Lab, University of Edinburgh & LIP6, Sorbonne Université / CNRS; originated at Nobel Symposium 210, 2026.

[30] Matthew L. Goh, Martin Larocca, Lukasz Cincio, M. Cerezo, and Frédéric Sauvage. Lie-algebraic classical simulations for quantum computing. *Physical Review Research*, 7:033266, 2025.

[31] Sarah Montgomery-Taylor, Hamsa Subramaniam, Rob Brisk, Liam Foster, Jack Ching, Andrew Marshall, Jim Taylor, John Hernandez, Jen McKay, Dillon Obika, Nnamdi Ezeanochie, Nira Goren, Tiffany Guo, Rachel Kwon, Arlene Chung, Jacqueline Shreibati, Stella Yeoum, Brett Hatfield, Zelda Gulledge, Ankit Nathu, Ferenc Kazinczi, Yun Liu, Jonathan Krause, Heather Cole-Lewis, Susan Thomas, Megan Jones Bell, Bakul Patel, Joelle Barral, Katherine Chou, Yossi Matias, Dana Gelb Safran, Elizabeth A. McGlynn, and Michael Howell. Towards multi-method evaluation of AI in health: Integrating benchmarks, autoraters, and expert evaluation. *arXiv preprint arXiv:2611.05127*, 2026.

[32] NHS England. DCB0129: Clinical risk management: its application in the manufacture of health IT systems. Technical report, NHS Digital / NHS England, 2018.

[33] NHS England. DCB0160: Clinical risk management: its application in the deployment and use of health IT systems. Technical report, NHS Digital / NHS England, 2018.

[34] Andrew Clegg, Chris Bates, John Young, Ronan Ryan, Linda Nichols, Elizabeth Ann Teale, Mohammed A. Mohammed, John Parry, and Tom Marshall. Development and validation of an electronic frailty index using routine primary care electronic health record data. *Age and Ageing*, 45(3):353–360, 2016.

[35] Rupert A. Payne, Silvia C. Mendonca, Marc N. Elliott, Catherine L. Saunders, Duncan A. Edwards, Martin Marshall, and Martin Roland. Development and validation of the Cambridge Multimorbidity Score. *Canadian Medical Association Journal*, 192(5):E107–E114, 2020.

[36] Feifei Bu, Alexandra Burton, Naomi Launders, Amelia E. Taylor, Alvin Richards-Belle, Stephanie Tierney, David Osborn, and Daisy Fancourt. National roll-out of social prescribing in England's primary care system: a longitudinal observational study using Clinical Practice Research Datalink data. *The Lancet Public Health*, 10(11):e903–e911, 2025.

[37] Quantinuum and SoftBank Corp. Quantum computing enterprise roadmap and fault-tolerant architecture white paper. Technical report, Quantinuum Ltd., 2026. Industry Strategic White Paper, July 2026.

[38] Jacob S. Nelson, Andrew J. Landahl, and Andrew D. Baczewski. Below-threshold execution of fault-tolerant ripple-carry addition on a trapped-ion quantum computer. *arXiv preprint arXiv:2610.14202*, 2026.

[39] Timothy Proctor, Oliver Hart, Oliver Widzowski Maupin, Matthew Girling, Daniel Hothem, Daniel Mills, Jordan Hines, Karl Mayer, Jacob S. Nelson, Tyler LeBlond, Zohim Chandani, Diego Forlivesi, Piper C. Wysocki, Boldizsár Poór, Joan M. Dreiling, Annie Park, Adam P. Reed, Brian Estey, Cameron Foltz, Akhil Isanaka, M. S. Allman, Michael Mills, Maxwell D. Urmey, Peter E. Siegfried, Audrey Faricy, Jin-Sung Kim, Cristina Cîrstoiu, Andrew D. Baczewski, Charles H. Baldwin, and Robin Blume-Kohout. Benchmarking the computational power of quantum computers. *arXiv preprint arXiv:2609.12146*, 2026.

[40] Austin G. Fowler, Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. Surface codes: Towards practical large-scale quantum computation. *Physical Review A*, 86(3):032324, 2012.

[41] Daniel Litinski. A game of surface codes: Large-scale quantum computing with lattice surgery. *Quantum*, 3:128, 2019.

[42] András Gilyén, Yuan Su, Guang Hao Low, and Nathan Wiebe. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, pages 193–204, 2019.

[43] Gilles Brassard, Peter Hoyer, Michele Mosca, and Alain Tapp. Quantum amplitude amplification and estimation. *Contemporary Mathematics*, 305:53–74, 2002.

[44] Ashley Montanaro. Quantum speedup of branch-and-bound algorithms. *Physical Review Research*, 2(1):013056, 2020.

[45] Jean-Luc Fauchère and Vladimir Pliška. Hydrophobic parameters π of amino-acid side chains from the partitioning of N-acetyl-amino-acid amides. *European Journal of Medicinal Chemistry*, 18(4):369–375, 1983.

[46] Matthew G. Durrant, Nicholas T. Perry, James J. Pai, Aditya R. Jangid, Januka S. Athukoralage, Masahiro Hiraizumi, John P. McSpedon, April Pawluk, Hiroshi Nishimasu, Silvana Konermann, and Patrick D. Hsu. Bridge RNAs direct programmable recombination of target and donor DNA. *Nature*, 630(8018):984–993, 2024.

[47] Marco Cerezo, Akira Sone, Tyler Volkoff, Lukasz Cincio, and Patrick J. Coles. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications*, 12(1):1791, 2021.